

An interval evolutionary algorithm based on dynamic relation adjustment strategy for many-objective problems

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ABSTRACT

Interval many-objective optimization problems (IMaOPs) are a kind of uncertain optimization problems. It is characterized by four or more objectives, at least one of which is an interval objective caused by parameter uncertainty. The upper and lower bounds of the interval objective make it difficult to take into account the relationship between convergence, diversity and uncertainty in the process of solving IMaOPs. In this paper, an interval many-objective evolutionary algorithm based on dynamic relation adjustment (IMaOEA-DRA) is proposed to aim at this challenge. Firstly, a dynamic interval pareto dominance (DIP-dominance) strategy is designed to adaptively adjust the proportion weight between individual uncertainty and convergence according to the number of iterations in the process of environment selection, so that the algorithm can maintain a good evolutionary direction while maintaining uncertainty. In addition, to ensure population diversity, penalty-based interval boundary intersections (PIBI) indicator is proposed and calculated in different clustering subspaces to select elite interval individuals. Finally, IMaOEA-DRA is tested on 23 benchmarks of IMaOPs. The numerical results show the superior performance of the proposed interval optimization method.

1. Introduction

Due to the inherent limitations of data, measurement conditions, and system noise, uncertain optimization problems are prevalent in real-world scenarios [1–4]. Interval many-objective optimization problems (IMaOPs), as a specific type of uncertainty optimization problem, combine the characteristics of both uncertainty optimization and many-objective optimization [5]. IMaOPs employ intervals to represent variable uncertainties, which enables obtaining upper and lower bounds with minimal information. Consequently, this approach exhibits notable convenience and efficiency in uncertainty modeling for various applications, such as resource scheduling [6–8], mechanical design [9–11], investment portfolio optimization [12–14], etc. Compared to stochastic planning and fuzzy planning, IMaOPs does not necessitate a substantial amount of uncertain information for constructing precise probability distributions or fuzzy membership functions of variables; therefore, IMaOPs exhibits enhanced universality in practical engineering problems characterized by the difficulty in obtaining uncertainty information. This paper primarily focuses on the solving method of IMaOPs

caused by parameter uncertainty, that is, interval many-objective evolutionary algorithms (IMaOEAs). Recently, transformation techniques (i.e. sampling [15], weighting [16], and mapping [17]) have been employed in IMaOEAs to convert the IMaOPs into MaOPs, and then existing deterministic optimization methods can be utilized to solve IMaOPs [18]. However, when the span of the interval is too large, the application of transformation techniques may inadvertently disrupt the inherent conflict relationship between objectives; hence, it is imperative to develop specific strategies (i.e. interval ordinal [19], reference intervals [20] and interval confidence level [21]) that can evaluate the intervals individual advantages and disadvantages intuitively, so as to comprehensively consider different aspects of the problem. By doing so, we can obtain a trade-off solution that aligns more closely with the characteristics of the problem and offers greater advantages in IMaOPs.

The above direct evaluation methods still exhibit limitations in IMaOPs. With respect to the IMaOPs, each uncertain objective is characterized by two parameters: an upper bound value and a lower bound value. An interval individual is dominant only if both the upper and lower bounds are smaller than other individuals; hence, as the number

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Table 1
Symbol definitions.

Notation	Explanation
Problem parameters:	
$f_i(x, u_i)$	lower bound of the i -th objective function
$f^-_i(x, u_i)$	upper bound of the i -th objective function
u_i	uncertainty vector parameter on the i -th objective
u_{ik}	the k -th component of u_i
$\underline{u}_{ik}$	lower bound of u_{ik}
$\bar{u}_{ik}$	upper bound of u_{ik}
D	decision variable space
R^n	n -dimensional space of real numbers
l	the dimension of u_i
M	number of objective functions
Algorithm parameters:	
N	population size
$maxFEs$	max function evaluations
α	rate of uncertainty decay
W	reference point matrix
λ_i	the i -th reference vector
S_a	population sorted by ascending order
$S_a[n]$	individuals in S_a
S^{DIP}_t	population selected by DIP-dominance ranking
S^{PIBI}_t	population selected by PIBI-indicator
F	pareto front
$F[k]$	set of individuals in the k -th pareto front layer
C_i	the i -th cluster
P	current population
U_i	uncertainty of the i -th individual

of objectives or the level of uncertainty within them increases, employing direct evaluation methods may lead to a disappearance of selection pressure and pose challenges in selecting effective candidate solutions [22–24].

The aim of IMaOEAs is to obtain trade-off solutions between diversity, convergence, and uncertainty [25]. The convergence ensures that the population evolves in the right direction; the diversity guarantees the dispersion of the solution set, while uncertainty reflects the stability of individuals. In the early stage of the algorithm, the individual with greater uncertainty has a larger interval hypervolume in the objective space, thereby increasing their likelihood of containing solutions with strong convergence. Consequently, these uncertain individuals effectively guide the population evolution toward optimal directions [26]. However, in the later stage of the algorithm, in order to seek solutions with better stability, it becomes crucial to deliberately select some individuals with lower uncertainty; meanwhile, this pursuit for stability inevitably necessitates sacrificing certain individuals that possess good convergence but exhibit high levels of uncertainty. In addition, diversity, as a group evaluation indicator, should be considered throughout the algorithm iteration process [27].

Based on the above viewpoints, we proposed an interval many-objective evolutionary algorithm based on dynamic relation adjustment (IMaOEA-DRA) which devises the dynamic adjustment strategies during environmental selection. This algorithm is tested on 23 benchmarks of IMaOPs and the numerical results demonstrate that IMaOEA-DRA offers further advantages in solving IMaOPs.

The contributions of this paper are as follows:

- A novel dynamic interval pareto dominance (DIP-dominance) strategy is proposed to dynamically adapt the convergence-uncertainty relationship among interval individuals during the iterative process of the algorithm, utilizing a decay function.
- A novel penalty-based interval boundary intersections (PIBI) indicator, utilizing subspace clustering, is proposed to evaluate the diversity of interval population throughout the iterations. Significantly, the enhancement of the PIBI indicator is achieved through the implementation of penalty-based boundary intersections (PBI) [28] in MaOPs.

The organization of this paper is as follows: the related works of using evolutionary algorithms to solve IMaOPs are summarized in Section 2, including transformation methods and direct evaluation methods. Section 3 first introduces the basic framework of IMaOEA-DRA, then describes and analyses the DIP-dominance, and PIBI indicator in detail. Parameter experiment, contrast experiments and numerical analysis are performed in Section 4. In Section 5, a conclusion is given.

2. Background study

In the uncertain problems of practical engineering, due to the limited prior information or the high cost of information collection, only the upper and lower bounds of parameters can be given; In such scenarios, IMaOEA holds significant value [29]. It is a widely recognized and effective approach for addressing IMaOPs within the framework of evolutionary algorithms.

The notations used in this article and their interpretations are provided in Table 1.

2.1. Problem statement

In general, an IMaOP [30] which problem parameters is interval form is defined as Eq. (1).

$$\begin{aligned}
 \min f &= (f_1(x, u_1), f_2(x, u_2), \dots, f_M(x, u_M))^T \\
 s.t. x &\in D \subseteq R^n, \\
 u_i &= (u_{i1}, u_{i2}, \dots, u_{il})^T, i = 1, 2, \dots, M, \\
 u_{ik} &= [\underline{u}_{ik}, \bar{u}_{ik}], k = 1, 2, \dots, l
 \end{aligned} \quad (1)$$

Here, x is an n -dimensional decision variable, and u_i is an l -dimensional uncertainty vector parameter, which is independent of the x . Obviously, if $u_{ik} = [\underline{u}_{ik}, \bar{u}_{ik}]$ is not considered, when $m = 1$, the problem reduces to a single-objective optimization problem in the general sense.

In IMaOPs, u_{ik} is represented by interval numbers $[\underline{u}_{ik}, \bar{u}_{ik}]$. Consequently, $f_i(x, u_i) \triangleq [f_i(x, u_i), f^-_i(x, u_i)]$, representing the i -th objective function, where $f_i(x, u_i)$ is the lower bound and $f^-_i(x, u_i)$ is the upper bound of optimization function $f_i(x, u_i)$, respectively.

For ease of understanding, we consider a simple multi-objective optimization problem with 2-dimensional decision variables and an uncertainty vector parameter, as Eq. (2).

$$\begin{aligned}
 \min_x f &= (f_1(x, u_1), f_2(x, u_2))^T \\
 f_1(x, u_1) &= -(x_1 \cdot u_{11} + x_2 \cdot u_{12}) \\
 f_2(x, u_2) &= x_1^2 + x_2^2 + u_{21} \cdot (x_1 + x_2)
 \end{aligned} \quad (2)$$

Here, $x = (x_1, x_2)$, $u_1 = (u_{11}, u_{12})^T$, $u_2 = u_{21}$. Notably, x_1 and x_2 are the components of the decision variable x , different from the decision variables x_i and x_j in the following text. At this point, the size of u_1 is 2×1 , and the size of u_2 is 1×1 . When u_i is expressed in terms of interval numbers, we obtain $u_{11} = [\underline{u}_{11}, \bar{u}_{11}]$, $u_{12} = [\underline{u}_{12}, \bar{u}_{12}]$, and $u_{21} = [\underline{u}_{21}, \bar{u}_{21}]$. Therefore, the corresponding objective function value are denoted as $f_1(x, u_1) \triangleq [f_1(x, u_1), f^-_1(x, u_1)]$, and $f_2(x, u_2) \triangleq [f_2(x, u_2), f^-_2(x, u_2)]$.

2.2. Related work

Traditional evolutionary algorithms are not well-suited for IMaOPs. This section will present relevant research from two perspectives: transformation-based methods and direct evaluation-based methods.

Transformation-based methods convert interval optimization problems into deterministic optimization problems using specific techniques. Gong et al [31]. successfully transformed the interval optimization problem into a deterministic bi-objective problem, considering the hypervolume and uncertainty of individuals. They further devised a set-based genetic algorithm to effectively solve this transformed deterministic optimization problem. In another study, Kumar et al [32]. introduced the concept of effective solutions obtained through different

Algorithm 1
 IMaOEA-DRA.

```

Input:  $N, M, \max FEs, \alpha$ ;
Output:  $P$ ;
1.  $P = \text{Initialization}(N)$ ;
2.  $W = \text{Reference point generation}(N, M)$ ;
3. while  $FEs < \max FEs$  do
4.    $\text{Mating pool} = \text{Mating selection}(P)$ ;
5.    $Q = \text{Offspring generation}(\text{Mating pool})$ ;
6.    $S \leftarrow P \cup Q$ ;
7.    $S^{DIP} = \text{DIP-dominance ranking}(S, \alpha, N) \rightarrow \text{see algorithm 2}$ ;
8.    $C = \text{clustering}(S, W) \rightarrow \text{see algorithm 3}$ ;
9.    $S^{PIBI} = \text{PIBI-indicator}(S, W) \rightarrow \text{see algorithm 4}$ ;
10.   $P = S^{PIBI}$ ;
11. end while

```

preference weights and utilized the weighted sum scalarization method in multi-objective programming to transform the interval optimization problem into a general deterministic nonlinear optimization problem. Moreover, In the truss static and dynamic topology optimization problem with interval parameter. Xu et al [33]. transformed the uncertain optimization problem by minimizing maximum standards and interval possibility, which was then solved using a genetic algorithm based on Pareto-dominance ranking.

In addition, Huang et al [34]. utilized an incremental shift vector technique to convert the uncertain nested optimization problem into an efficient sequential iterative process involving deterministic design optimization and hybrid reliability analysis. Jin et al [35]. tackled the uncertain optimization problem in electromagnetic track launchers by utilizing interval order relations for performance optimization and reliability analysis. Furthermore, Li et al [36]. transformed the interval optimization problem using ordinal numbers and probability degrees of intervals, while conducting sensitivity tests to analyse the impact of probability degree and weight coefficient on objectives. Similarly, Fu et al [37]. employed the satisfaction value of interval possibility degree to effectively select individuals within the framework of a differential evolution algorithm. Notably, they integrated dimension reduction interval analysis to transform the nested optimization problem into a single-loop optimization problem.

The advantage of the above methods is that the interval is reformulated in the form of sets, thereby streamlining the solution process. Nevertheless, these methods may result in a loss of the inherent conflict relationship between objectives in IMaOPs, leading to inaccuracies in the final outcome [38].

In contrast, direct evaluation-based methods are also widely adopted for interval optimization problems, primarily emphasizing the refinement of interval evaluation strategies. Costa et al [39]. introduced a redefinition of vector space, order relation, and algebraic calculus within the framework of interval space. They also developed problem representation and calculation methods that are tailored to this context. Na et al [40]. proposed an interval particle swarm optimization algorithm and utilized the satisfaction value of interval possibility degree for selecting individuals in the population. Zhang et al [41]. devised an innovative interval micro-immune optimization method based on possibility degree. Initially optimizing the interval boundary threshold allowed for subsequent collection of optimal solutions from valid alternatives.

Based on their research, Karmakar et al [42]. divided the interval space into several different subregions and computed the interval objective function value in each feasible subregion; subsequently, an interval ranking rule for the pessimistic decision maker view was

applied to select individuals. Similarly, Fu et al. [43] proposed an interval differential evolution method based on adaptive subinterval decomposition analysis to directly solve the nonlinear uncertain optimization problem with interval parameters.

For multi-objective optimization problems, the consideration of convergence, diversity, and uncertainty trade-offs become crucial when contemplating direct evaluation strategies [44]. Liu et al. [45] proposed an algorithm for interval multi-objective optimization that incorporates a multidimensional parallelepiped interval model. This model effectively captures the uncertainty through the use of marginal intervals and describes the degree of interval dependence using correlation coefficients.

Cui et al [46]. developed an alternative approach to solve interval optimization problems directly by introducing an elite genetic strategy in their algorithm for interval multi-objective optimization. Their method incorporates an interval confidence dominance relation and utilizes the concept of interval crowding distance. Guo et al [47]. divided IMOPs into multiple subproblems based on weight and enhanced the Chebyshev aggregation function to accommodate interval calculations within the framework of MOEA/D.

In summary, while the direct evaluation-based methods can mitigate errors caused by problem transformation, the primary challenge in solving interval optimization problems within the evolutionary computation framework lies in selecting appropriate interval individuals. IMaOPs differ from traditional many-objective problems in that uncertainty is introduced as an evaluation criterion when an individual is extended to an interval representation. However, existing individual selection schemes fall short in considering the dynamic equilibrium between convergence, diversity, and uncertainty [48–50]. To tackle these challenges, IMaOEA-DRA was developed by meticulously analysing convergence, diversity, and uncertainty throughout the population iteration process. Notably, it incorporates concepts from direct evaluation-based methods to circumvent errors resulting from problem transformation.

3. Interval many-objective evolutionary algorithm based on dynamic relation adjustment

In this paper, an evolutionary algorithm framework is used to solve the IMaOPs. The proposed IMaOEA-DRA represents a significant advancement over the conventional evolutionary algorithm framework, and its enhanced structure is presented in Algorithm 1.

The fundamental concept of IMaOEA-DRA lies in dynamically adjusting the interplay between convergence, diversity, and uncertainty throughout the iterative process. Specifically, their relationship adheres

Algorithm 2

DIP-dominance ranking.

Input: S, α, N ;
Output: S_t^{DIP} ;

1. $F = []$;
2. $S_a \leftarrow$ ascending order of the objective (S);
3. **for** every individual $S_a[n]$ **in** S_a
4. $count = \text{size}(F)$; {the number of fronts already assigned}
5. $k = 1$;
6. $sorted = 0$;
7. **while** $sorted < N$ **do**
8. calculate U_i with eq. (5) and (7);
9. compare individuals in $S_a[n]$ and $F[k]$ with eq. (6);
10. **if** $F[k]$ have no individual dominating $S_a[n]$
11. $F[k] \leftarrow S_a[n]$;
12. **break**;
13. **else**
14. $k = k + 1$;
15. **if** $k > count$
16. $F[k] \leftarrow S_a[n]$;
17. **break**;
18. **end if**
19. **end if**
20. $sorted = \text{size}(F[k])$; {The number of sorted individuals}
21. **end while**
22. **end for**
23. $S_t^{DIP} \leftarrow F[k]$;

Algorithm 3

Clustering.

Input: S_t^{DIP}, W ;
Output: C ;

1. $\lambda_i, i \in \{1, 2, \dots, N\} \leftarrow$ reference vectors generate (W);
2. $\{C_1, C_2, \dots, C_N\} \leftarrow$ cluster generate ();
3. **for** each x **in** S_t^{DIP} **do**
4. $d_{min} \leftarrow d_{1,22}$;
5. **for** each j **in** 2 to n
6. **if** $d_{j,22} < d_{min}$
7. $d_{min} \leftarrow d_{j,22}$;
8. $C_i \leftarrow x_j, i \in \{1, 2, \dots, N\}$;
9. **end if**
10. **end for**
11. **end for**

to the following principles: during the early iterations of the algorithm, emphasis is primarily placed on convergence; as the number of iterations increases, there is a proportional increase in uncertainty; and diversity remains a constant consideration across all iterations. IMAOEA-DRA achieves the above concept through the utilization of the DIP-dominance strategy, clustering strategy, and PIBI indicator.

Step 1 randomly generates the initial population according to the pre-set population size. Step 2 uniformly generates reference points in the objective space according to [51], which are applied in the environmental selection to divide the objective space (see Section 3.2). If the $maxFEs$ has not been reached, the algorithm executes binary tournament

selection and standard genetic operators to generate offspring in Steps 4–5. Then, P and Q are merged to prepare for DIP-dominance (see Algorithm 2) in Step 6. The DIP-dominance strategy dynamically adjusts the relationship between uncertainty and convergence by implementing a decay function, which will be elaborated upon in Section 3.1. Step 8 executes the clustering operations, producing the cluster subspace (see Algorithm 3), and Step 9 executes the PIBI-indicator (see Algorithm 4) to ensure diversity is maintained. Notably, the clustering strategy remains the same as in θ -DEA [52].

Algorithm 4
PIBI indicator.

Input: C, N ;
Output: S_i^{PIBI} ;

1. **for** each i in 1 to N
2. **for** each x in C_i
3. d_{PIBI} = calculate PIBI distance with equation (9);
4. $ranking$ = sort (d_{PIBI});
5. **end for**
6. **end for**
7. **for** each i in 1 to N
8. $S_i^{PIBI} \leftarrow$ selecting the ranking (top 1) in C_i ;
9. $ranking$ (top 1) $\leftarrow$ inf;
10. **if** size (S_i^{PIBI}) = N
11. **break**;
12. **end if**
13. **end for**

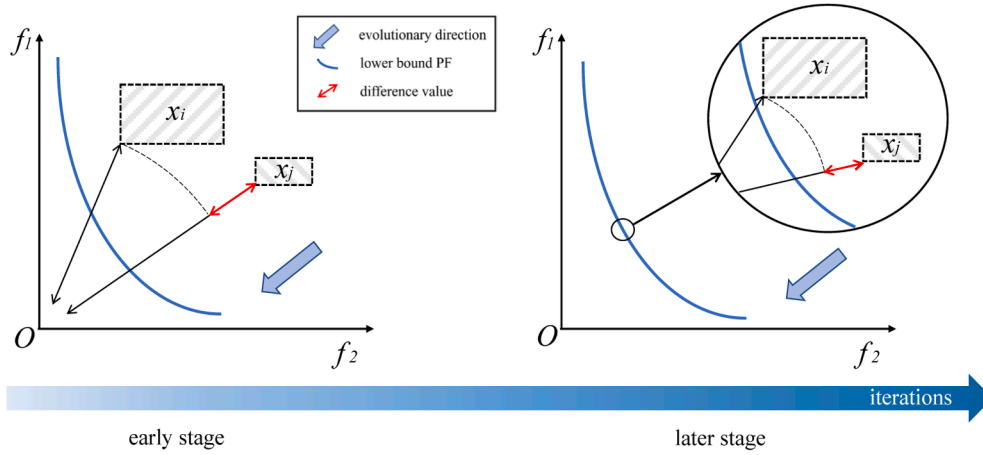


Fig. 1. The interval individuals are in two-dimensional space.

3.1. Dynamic interval pareto dominance ranking

Firstly, given two feasible solutions x_i and x_j , the Pareto dominance is defined by [53] as *Definition 1*; here, $f(x_i)$ and $f(x_j)$ are the fitness values for feasible solutions x_i and x_j , respectively. Pareto dominance is a classical method in MaOPs.

Definition 1 (Pareto-dominance): x_i dominates x_j , denoted by $x_i \prec x_j$, iff (1) $\forall p \in \{1, 2, \dots, M\}$, $f_p(x_i)$ is no more than $f_p(x_j)$; (2) $\exists q \in \{1, 2, \dots, M\}$, $f_q(x_i)$ is less than $f_q(x_j)$, as Eq. (3). If the above condition is not satisfied, denoted as $x_i \parallel x_j$, it means that x_i and x_j do not dominate each other.

$x_i \prec x_j \Leftrightarrow$

$$\begin{cases} \forall p \in \{1, 2, \dots, M\}, (f_p(x_i) \leq f_p(x_j)) \wedge (x_i \neq x_j); \\ \exists q \in \{1, 2, \dots, M\}, (f_q(x_i) < f_q(x_j)) \wedge (x_i \neq x_j); \end{cases} \quad (3)$$

In the interval form, for Eq. (1), Given two feasible solutions x_i and x_j , whose corresponding objective function values are $[f(x_i, u), \bar{f}(x_i, u)]$, and $[f(x_j, u), \bar{f}(x_j, u)]$. According to the interval ordinal relation, the interval pareto dominance (IP-dominance) between $f(x_i, u)$ and $f(x_j, u)$ is defined as follows [19].

Definition 2 (IP-dominance): x_i IP-dominates x_j , denoted by $x_i \prec_{IP} x_j$, iff (1) $\forall p \in \{1, 2, \dots, M\}$, $f_p(x_i, u_p)$ is less than $f_p(x_j, u_p)$ in the interval sense, or $f_p(x_i, u_p)$ and $f_p(x_j, u_p)$ are incomparable in the interval sense;

(2) $\exists q \in \{1, 2, \dots, M\}$, $f_q(x_i, u_q)$ is less than $f_q(x_j, u_q)$ in the interval sense, as Eq. (4). If the above condition is not satisfied, denoted as $x_i \parallel_{IP} x_j$, it means that x_i and x_j do not dominate each other.

$x_i \prec_{IP} x_j \Leftrightarrow$

$$\begin{cases} \forall p \in \{1, 2, \dots, M\}, (f_p(x_i, u_p) <_{IN} f_p(x_j, u_p)) \\ \vee (f_p(x_i, u_p) \parallel_{IN} f_p(x_j, u_p)) \wedge (x_i \neq x_j); \\ \exists q \in \{1, 2, \dots, M\}, (f_q(x_i, u_q) <_{IN} f_q(x_j, u_q)) \wedge \\ (x_i \neq x_j); \end{cases} \quad (4)$$

Remark 1. the notation " $<_{IN}$ " and " $\parallel_{IN}$ " are "less than" and "incomparable" in the interval sense. $f(x_i, u) <_{IN} f(x_j, u)$ means that $f(x_i, u)$ is no more than $f(x_j, u)$, and $\bar{f}(x_i, u)$ is no more than $\bar{f}(x_j, u)$, and $f(x_i, u) \neq f(x_j, u)$. If $f(x_i, u) <_{IN} f(x_j, u)$ and $f(x_j, u) <_{IN} f(x_i, u)$ do not hold, $f(x_i, u)$ and $f(x_j, u)$ are incomparable, denoted as $f(x_i, u) \parallel_{IN} f(x_j, u)$.

However, the satisfaction condition of IP-dominance is slack, that is, even if the objective does not rise to a high dimension, the number of non-dominated solutions will greatly increase, which brings difficulties to environment selection; thus, individual uncertainty is considered in this paper.

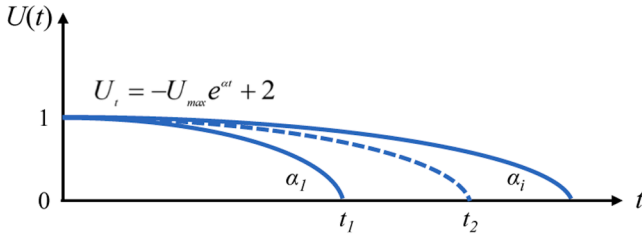


Fig. 2. The decay function.

Fig. 1 takes a two-dimensional space as an example to illustrate the relation of two individuals in early and later stages of the algorithm. For x_i , despite its initial poor uncertainty in the algorithm, its convergence surpasses that of x_j . Hence, we are inclined to select it for accelerating the convergence of algorithm. However, as the individuals approach the PF during the later stages of the algorithm, convergence advantage (difference value) of x_i becomes fewer apparent. Consequently, we prefer opting for individuals with higher uncertainty levels. Therefore, we expect that the dominant comparison of individuals in the early and late stages of the algorithm can consider different uncertainty thresholds, respectively.

Concretely, in the early stage, the algorithm pays attention to the convergence, which individuals with large uncertainty and good convergence will be selected to promote the evolution of the population in the direction of more advantages, that is, algorithm dominated by convergence, and has a relatively large tolerance for individual uncertainty; in the later stage, the algorithm pays more attention to the uncertainty of the individual, and some individuals with poor convergence but good uncertainty will be selected to ensure the stability of the solution set; in the later stage of the algorithm, that is, the algorithm dominated by uncertainty, weakening the role of convergence, so as to obtain a more stable solution. It is worth noting that in terms of convergence, the algorithm performs the calculation of the indicator based on decomposition strategy, so as to ensure population convergence.

Therefore, this paper proposes a dynamic interval pareto dominance (DIP-dominance) based on pareto dominance and IP-dominance. Firstly, an additional condition (see Fig. 3) is imposed on IP-dominance, and then an uncertainty decay function is designed (see Eq. (7)). Here, the pareto dominance between x_i and x_j on its lower bound objective is defined as $x_i \prec_l x_j$, and the pareto dominance on its upper bound objectives is defined as $x_i \prec_u x_j$.

$$U_i = \sqrt{\frac{(\overline{f}_1(x_i, u_i) - \underline{f}_1(x_i, u_i))^2 + (\overline{f}_2(x_i, u_i) - \underline{f}_2(x_i, u_i))^2 + \dots + (\overline{f}_m(x_i, u_m) - \underline{f}_m(x_i, u_m))^2}{m}} \quad (5)$$

Additionally, the uncertainty is measured by the Eq. (5). Where U_i represents the interval diagonal distance of individual x_i in a high dimensional space.

Definition 3(DIP-dominance). Given two feasible solutions x_i and x_j , x_i DIP-dominates x_j , defined $x_i \prec_{DIP} x_j$, iff x_i and x_j satisfy the following condition (6).

$$\begin{aligned} x_i \prec_{DIP} x_j \Leftrightarrow & \{x_i \prec_l x_j \wedge x_i \prec_u x_j\} \vee \{x_i \prec_l x_j \wedge x_i \succ_u x_j\} \vee \\ & \{x_i \prec_l x_j \wedge x_i \parallel_u x_j\} \vee \{x_i \succ_l x_j \wedge x_i \prec_u x_j\} \vee \\ & \{x_i \succ_l x_j \wedge x_i \parallel_u x_j\} \vee \{x_i \parallel_l x_j \wedge x_i \prec_u x_j\} \vee \\ & \{x_i \parallel_l x_j \wedge x_i \succ_u x_j\} \vee \{x_i \parallel_l x_j \wedge x_i \parallel_u x_j\} \wedge \\ & \{U_i < U_j\} \wedge \{U_i < U_t\} \end{aligned} \quad (6)$$

The U_t is a threshold value of uncertainty that varies with t , which calculated by Eq. (7).

$$U_t = -U_{max} e^{\alpha t} + 2 \quad (7)$$

Where α is a decay parameter of U_t , and U_{max} is the max U_i of the population in the 1-th iteration. In the first generation of the population without iteration, the individual with the largest uncertainty is also roughly the largest during the whole algorithm iteration. This is used as a baseline for attenuation. In order for U_t to vary in a range that always does not exceed itself, we give a range of α , as follows in Eq. (8).

$$0 = -U_{max} e^{\beta t} + 2 \Leftrightarrow \beta = \frac{1}{t_{max}} \ln \left(\frac{2}{U_{max}} \right) \quad (8)$$

Here, the range of α is $[0, \beta]$. The more α tends to 0, the smoother the decay curve in the early iteration; while α tends to β , the steeper the decay curve in the late iteration, as shown in Fig. 2, and the parameter sensitivity experiment is set in Section 4.

In Fig. 3, considering two individuals x_i and x_j , (a)-(b) denotes the upper bound values are dominated by x_j , non-dominated, and dominating x_j , when the lower bound value of x_i dominates the lower bound value of x_j .

(d)-(f) represent the upper bound values are dominated by x_j , non-dominated, and dominating x_j , when the lower bound value of x_j dominates the lower bound value of x_i . (g)-(i) show the upper bound values are non-dominated, dominated by x_j , and dominating x_j , when the lower bound value of x_i and x_j are non-dominates.

The relationships between interval individuals are diverse. In DIP-dominance, we determine whether two individuals are dominated by the relationship between upper bound value, lower bound value, and uncertainty. This operation can relax the interval dominance relationship to a certain extent. Next, we will prove that DIP-dominance is irreflexive, asymmetric, and transitive.

Property 1. (Irreflexive): if $x \in D$ and x is a feasible solution, then $x \prec_{DIP} x$ is invalid.

Proof. Suppose that $x \prec_{DIP} x$, then $\exists q \in \{1, 2, \dots, M\}$, $\{f_q(x, u_q) \leq f_q(x, u_q)\} \wedge \{f_q(x, u_q) \leq f_q(x, u_q)\} \wedge \{x \neq x\}$ by Definition 3, however, $x = x$; thus $x \prec_{DIP} x$ is invalid.

Property 2. (asymmetric): if x_i and $x_j \in D$, and x_i and x_j are all feasible solution, satisfy $x_i \prec_{DIP} x_j$, then $x_j \prec_{DIP} x_i$ is invalid.

Proof. Suppose that $x_j \prec_{DIP} x_i$, then $U_j < U_i$, according Definition 3; however, satisfy $x_i \prec_{DIP} x_j$, $U_i < U_j$, thus, $x_j \prec_{DIP} x_i$ is invalid.

Property 3. (transitive): if x_i , x_j and $x_k \in S$, and x_i , x_j and x_k are all feasible solution, satisfy $x_i \prec_{DIP} x_j$, $x_j \prec_{DIP} x_k$, then $x_i \prec_{DIP} x_k$.

Proof. Here, we given the converse-negative proposition of Definition 3, that is, $\{x_i \succ_l x_j \wedge x_i \succ_u x_j\} \vee \{U_i \geq U_j\} \vee \{U_i \leq U_t\}$, iff $\{x_i \succ_{DIP} x_j\} \vee \{x_i \parallel_{DIP} x_j\}$. According to the converse-negative proposition and Definition 1, $x_i \succ_{DIP} x_j$, that is $\forall q \in \{1, 2, \dots, M\}$, $f_q(x_i, u_q) \geq f_q(x_j, u_q) \wedge f_q(x_i, u_q) \geq f_q(x_j, u_q) \wedge x_i \neq x_j$, or $U_i > U_j$, or $U_i > U_t$; $x_j \succ_{DIP} x_k$, that is $\forall q \in \{1, 2, \dots, M\}$, $f_q(x_j, u_q) \geq f_q(x_k, u_q) \wedge f_q(x_j, u_q) \geq f_q(x_k, u_q) \wedge x_j \neq x_k$, or $U_j > U_k$, or $U_j > U_t$, easy to know, $f_q(x_i, u_q) \geq f_q(x_j, u_q) \geq f_q(x_k, u_q) \wedge f_q(x_i, u_q) \geq f_q(x_k, u_q) \wedge x_i \neq x_k$, or $U_i > U_j > U_k$, thus, According to the converse-negative proposition of Definition 3, $x_i \succ_{DIP} x_k$.

DIP-dominance acts on the whole iterative process of the algorithm and imposes an additional condition on pareto dominance through the decay function to decrease the number of candidate solutions.

The algorithm flow of DIP-dominance ranking is shown in Algorithm 2. Algorithm 2 employs the DIP-dominance to ranking the individuals in the combined population. First, step 1 set an empty set F to hold the sorted individuals; then, when the number of individuals in F is less than the prescribed candidate population size N .

Steps 7–9 iteratively compare the combined population S_q and F with the DIP-dominance through Definition 2; finally, steps 10 and 20 put the individuals into F in the order of k layers. Notably, S_q is the first pre-

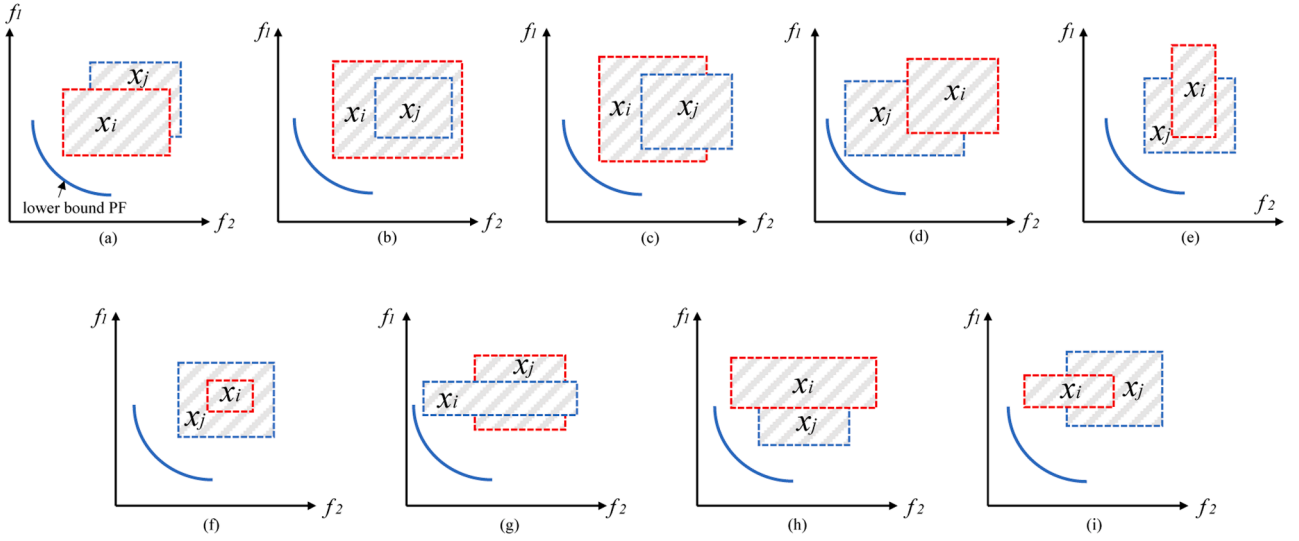


Fig. 3. The interval relation is in two-dimensional space.

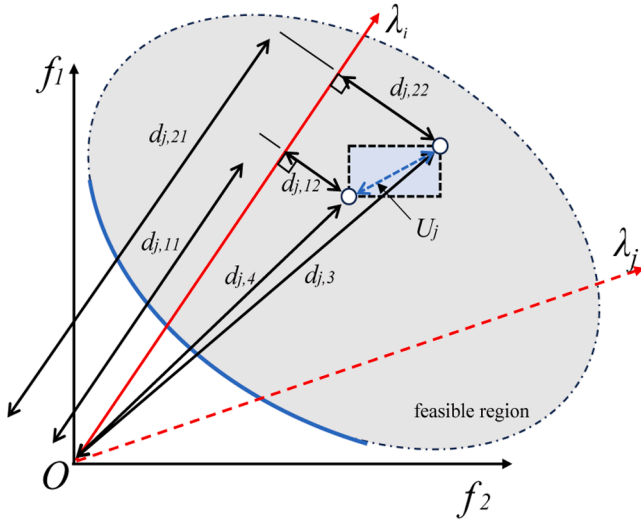


Fig. 4. The interval individuals and reference vectors.

Table 2
Characteristics of the test functions.

Characteristics	Test functions
multimodal	IDTLZ1, IDTLZ3, IWFG2, IWFG4, IWFG9, IMaF3–4
linear	IDTLZ1, IWFG3, IMaF1, IMaF8, IMaF9, IMaF14
concave	IDTLZ2–4, IWFG4–9, IMaF2, IMaF4, IMaF6, IDTLZ4, IWFG1, IWFG7–9, IMaF5
biased	IWFG2, IDTLZ7
disconnected	IWFG1–9
multiscale	IWFG2, IWFG3, IWFG6, IWFG8, IWFG9
nonseparable	IWFG3, IMaF6, IMaF8, IMaF9
degenerate	IWFG5, IWFG9
deceptive	IWFG2, IMaF3, IMaF5, IMaF15
convex	IWFG9
mixed	IMaF14, IMaF15
large scale & partially separable	

Table 3
Parameters setting.

Parameters	Value
N	100
M	4,5,6,8,10
D	8,9,10,12,14 (200 in large scale)
maxFEs	20,000
run times	30

sorted by the objective successively in step 2. The algorithm 2 is based on the framework of [54].

3.2. Clustering

According to the traditional evolutionary algorithm, the diversity is typically measured by the distance between individuals. However, due to the particularity of the interval population, when the uncertainty is large in the early stage of the algorithm, the interval individuals normally overlap with each other, and the diversity of the population cannot be measured by the distance between individuals.

In this section, the clustering operation is introduced to maintain the diversity of the interval population, which divides the objective space into multiple subspaces by clustering clusters, and the individuals in each subspace are regarded as a subpopulation for environment selection (see algorithm 3), and Fig. 4 is the relationship between interval individuals and reference vectors.

Where λ_i is the reference vector. For an interval individual, whether in two-dimensional space or high-dimensional space, the vertical distance between the lower bound value and λ_i is calculated, denoted as $d_{j,12}$, and the vertical distance between the upper bound value and λ_i is denoted as $d_{j,22}$. In clustering, step 2 firstly generates N empty cluster, where N is the population size. Ideally, we cluster S_t into N clusters and place them into $\{C_1, C_2, \dots, C_N\}$. In addition, we regard the smallest $d_{j,22}$ as the cluster center, because if $d_{j,22}$ is small, then $d_{j,12}$ must be small, without considering the uncertainty, so as to the individual is must the closest to the reference vector. This guarantees that there is no loss of diversity. $d_{j,11}$ and $d_{j,21}$ are distances that reflect convergence. Following this, Algorithm 3 choose the best individual from each cluster, aiming to maintain the diversity of the population by this method.

However, in practice, there may be empty sets in some clusters, especially when the dimension of the objective space increases, causing a large number of individuals to pile up in certain clusters.

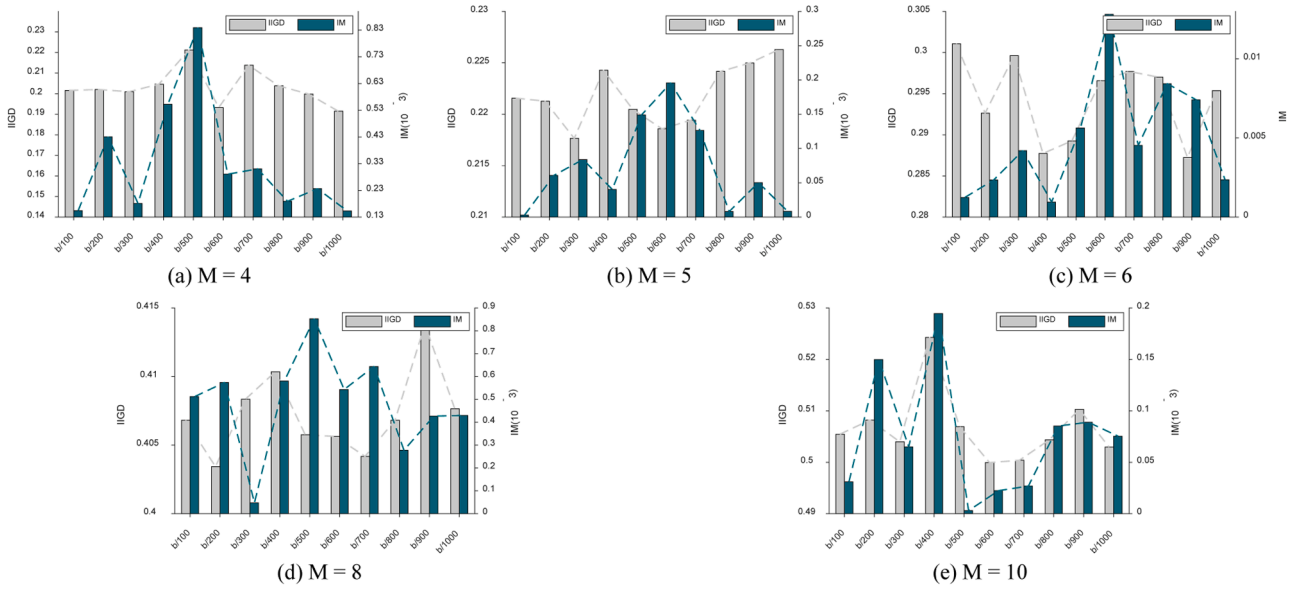
Fig. 5. IIGD and IM obtained across IDTLZ2 in $M = 4, 5, 6, 8, 10$.

Table 4

IIGD (mean) obtained across IDTLZ.

M	4				8				10			
Algorithms	I	P	D	DRA	I	P	D	DRA	I	P	D	DRA
IDTLZ1	0.145	0.080	Nan	0.055	0.743	0.176	Nan	0.289	0.739	0.199	Nan	0.276
IDTLZ2	0.169	0.157	0.425	0.140	0.463	0.395	0.758	0.399	0.548	0.499	0.814	0.389
IDTLZ3	Nan	0.440	Nan	0.934	6.408	1.927	Nan	1.071	Nan	2.163	Nan	1.896
IDTLZ4	0.178	0.217	0.593	0.148	0.539	0.494	1.627	0.412	1.005	0.595	1.595	0.476
IDTLZ7	20.30	5.861	13.89	7.949	42.17	25.18	30.41	26.74	44.38	41.86	38.70	41.29

Table 5

IHV (mean) obtained across IDTLZ.

M	4				8				10			
Algorithms	I	P	D	DRA	I	P	D	DRA	I	P	D	DRA
IDTLZ1	0.468	0.511	Nan	0.826	0.048	0.345	Nan	0.192	0.020	0.314	Nan	0.242
IDTLZ2	0.509	0.469	0.179	0.648	Nan	0.0061	Nan	0.126	0.447	0.460	0.123	0.551
IDTLZ3	Nan	0.269	Nan	0.151	0.042	0.390	0.097	0.501	Nan	0.068	Nan	0.171
IDTLZ4	0.516	0.464	0.078	0.628	0.399	0.479	Nan	0.572	0.058	0.488	Nan	0.546
IDTLZ7	Nan	7.1e-3	Nan	Nan	Nan	Nan	Nan	Nan	Nan	Nan	Nan	Nan

Table 6

IM (mean) obtained across IDTLZ.

M	4				8				10			
Algorithms	I	P	D	DRA	I	P	D	DRA	I	P	D	DRA
IDTLZ1	0.715	0.757	0.568	0.012	0.0141	2.8e-3	1.0e-2	8.7e-5	5.7e-4	1.0e-4	1.10e-3	4.3e-5
IDTLZ2	0.633	0.659	0.802	3.1e-3	0.0135	9.3e-4	7.3e-3	8.7e-4	1.2e-4	1.1e-4	9.5e-4	1.2e-4
IDTLZ3	0.859	0.847	0.520	0.396	6.77e-3	4.0e-3	9.5e-3	1.7e-3	2.7e-4	7.0e-5	7.8e-4	1.5e-4
IDTLZ4	0.578	0.637	0.934	1.2e-3	0.0311	1.2e-3	8.9e-3	1.0e-4	1.9e-3	7.6e-5	7.7e-4	1.1e-5
IDTLZ7	2.6e-3	1.45e-3	0.945	7.6e-8	6.3e-22	1.3e-7	9.4e-3	7.2e-20	2.8e-24	2.4e-22	9.6e-4	4.2e-28

In such cases, it is necessary to sort the individuals in the cluster using Algorithm 4, and then traverse C_1 to C_N to select the top individual from each cluster. If the population size is still not satisfied, we continue to select the next top individuals sequentially.

The distance between the first individual x and the nearest reference vector λ_i in the sorted set S_t^{DIP} is taken as initial d_{min} in step 4, and then the distance between the remaining individuals and the reference vector λ_i is calculated in loop. If the distance is less than the current d_{min} , the individual is classified into cluster i , and then the current d_{min} is replaced in step 6–10. The clustering ensures the convergence of population in

the evolutionary process.

3.3. The indicator of penalty-based interval boundary intersections

When the uncertainty of the dominant population is adjusted through the dominance relation, the PIBI indicator ensures that population diversity is maintained while preventing any movement towards increased uncertainty. The PIBI indicator is defined as Eq. (9).

$$d_{j,PIBI} = d_{j,11} + d_{j,21} + \theta(d_{j,12} + d_{j,22}) \quad (9)$$

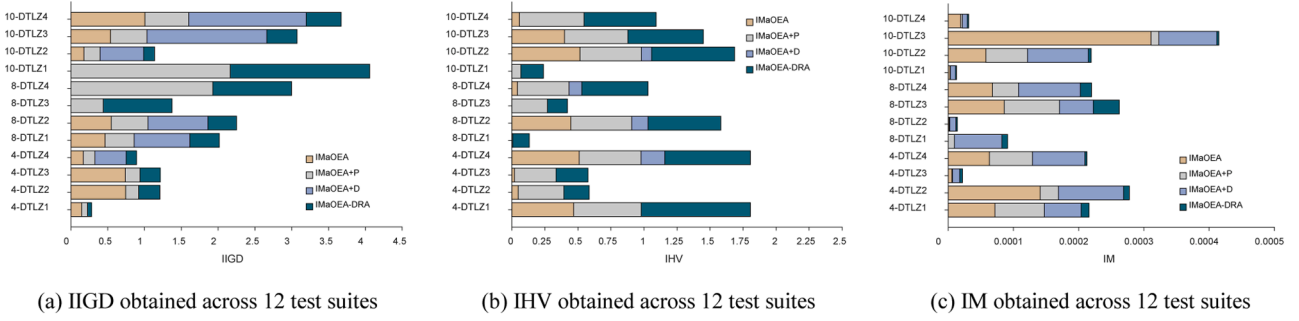


Fig. 6. Stacked bar of IIGD, IHV, IM with different strategies.

Table 7

IIGD (mean) and IHV (mean) obtained across IDTLZ 1–4.

Metric		IIGD					IHV					
Pro.	M	DI-μMOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA	DI-μMOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA	
IDT	4	1.1878e-1(-)	8.1832e-1(-)	9.5447e-2(-)	5.6092e-1(-)	5.5808e-2	2.1481e-1(-)	1.5852e-1(-)	5.4037e-1(-)	0.000e+0(-)	8.2632e-1	
	LZ1	5	4.5030e-1(-)	3.7102e+0(-)	1.2751e-1(-)	2.0597e-1(-)	9.0798e-2	5.0045e-1(-)	7.0253e-2(-)	5.1273e-1(-)	0.000e+0(-)	7.1595e-1
	6	2.0115e+0(-)	5.8066e+0(-)	1.6362e-1(=)	4.5939e+0(-)	1.6411e-1	3.9775e-1(-)	0.000e+0(-)	4.7372e-1(-)	0.000e+0(-)	5.3206e-1	
	8	3.1445e+0(-)	2.3210e+0(-)	2.4616e-1(=)	7.3798e+0(-)	2.8976e-1	0.000e+0(-)	0.000e+0(-)	3.6806e-1(+)	0.000e+0(-)	1.9227e-1	
	10	2.2129e+0(-)	8.3805e+0(-)	2.9794e-1(-)	4.7124e+0(-)	2.7629e-1	7.9791e-4(-)	0.000e+0(-)	2.9230e-1(+)	0.000e+0(-)	2.4216e-1	
IDT	4	3.9856e-1(-)	4.7312e-1(-)	2.5015e-1(-)	3.5706e-1(-)	1.4096e-1	0.000e+0(-)	0.000e+0(-)	4.8131e-1(-)	5.6276e-1(-)	6.4891e-1	
	LZ2	5	3.5315e-1(-)	8.0670e-1(-)	3.6234e-1(-)	8.1074e-1(-)	2.1266e-1	0.000e+0(-)	0.000e+0(-)	4.8530e-1(-)	6.0614e-2(-)	6.7466e-1
	6	6.8027e-1(-)	1.6851e+0(-)	4.3979e-1(-)	1.6110e+0(-)	2.8106e-1	0.000e+0(-)	0.000e+0(-)	4.9370e-1(-)	0.000e+0(-)	6.0316e-1	
	8	1.3918e+0(-)	1.6965e+0(-)	5.9336e-1(-)	1.8702e+0(-)	3.9954e-1	0.000e+0(-)	0.000e+0(-)	4.5663e-1(-)	0.000e+0(-)	5.0023e-1	
	10	1.2935e+0(-)	1.7994e+0(-)	7.1945e-1(-)	1.9066e+0(-)	4.8976e-1	0.000e+0(-)	0.000e+0(-)	4.2575e-1(-)	0.000e+0(-)	5.0053e-1	
IDT	4	2.2002e-1(-)	3.9859e+0(-)	2.0808e+0(-)	1.1335e-1(+)	9.3490e-1	0.000e+0(-)	0.000e+0(-)	9.3896e-2(-)	0.000e+0(-)	1.5057e-1	
	LZ3	5	6.6791e-1(+)	7.1902e+0(-)	2.1592e+0(-)	7.0896e+0(-)	1.0037e+0	0.000e+0(-)	0.000e+0(-)	8.2605e-2(-)	0.000e+0(-)	3.1519e-1
	6	2.2646e-1(+)	1.0951e+0(-)	1.7658e+0(-)	1.0451e+0(-)	8.2995e-1	0.000e+0(-)	0.000e+0(-)	1.1731e-1(-)	0.000e+0(-)	3.1490e-1	
	8	5.8117e+0(-)	1.3489e+0(-)	2.7175e+0(-)	1.2405e+0(-)	1.0719e+0	0.000e+0(-)	0.000e+0(-)	1.5235e-4(-)	0.000e+0(-)	1.2637e-1	
	10	7.7483e+0(-)	9.8915e+0(-)	2.4166e+0(-)	1.1741e+0(-)	1.0964e+0	0.000e+0(-)	0.000e+0(-)	7.9104e-2(+)	0.000e+0(-)	6.8492e-2	
IDT	4	2.1525e-1(-)	4.6460e-1(-)	5.4807e-1(-)	3.9881e-1(-)	1.4846e-1	5.3969e-1(-)	4.6327e-1(-)	4.0678e-1(-)	4.9359e-1(-)	6.2764e-1	
	LZ4	5	3.6237e-1(-)	7.7695e-1(-)	7.2285e-1(-)	6.4283e-1(-)	2.3094e-1	5.4857e-1(-)	6.4708e-2(-)	3.9548e-1(-)	4.5824e-1(-)	6.7119e-1
	6	5.2971e-1(-)	1.2442e+0(-)	6.2458e-1(-)	1.7119e+0(-)	3.1510e-1	4.0864e-1(-)	1.5697e-3(-)	4.5629e-1(-)	0.000e+0(-)	6.8096e-1	
	8	9.4335e-1(-)	1.7282e+0(-)	7.2687e-1(-)	1.4419e+0(-)	4.5278e-1	4.3879e-2(-)	0.000e+0(-)	4.6252e-1(-)	2.1626e-3(-)	5.7245e-1	
	10	1.9438e+0(-)	2.0974e+0(-)	8.0872e-1(-)	1.7002e+0(-)	5.7631e-1	0.000e+0(-)	0.000e+0(-)	4.3777e-1(-)	0.000e+0(-)	4.8858e-1	
+/-/=		2/18/0	0/20/0	0/18/2	1/19/0		0/20/0	0/20/0	3/17/0	0/20/0		

Here, θ is a predefined penalty parameter[52]. Next, we will prove that the PIBI distance reflects the convergence and diversity of the population without loss uncertainty.

Property 4. (no loss uncertainty): $U_j < d_j, \text{PIBI}$.

Proof. The PIBI distance is improved by the PBI distance; hence, the proof of the PBI distance will not be repeated. The triangle formed by $d_{j,4}$, $d_{j,3}$ and U_j is denoted as ΔA ; The triangle formed by $d_{j,11}$, $d_{j,12}$ and $d_{j,4}$ is denoted as ΔB ; similarly, the triangle formed by $d_{j,21}$, $d_{j,22}$ and $d_{j,3}$ is denoted as ΔC . Easy to know, in ΔA , $U_j < d_{j,4} + d_{j,3}$; in ΔB , $d_{j,4} < d_{j,11} + d_{j,12}$; in ΔC , $d_{j,3} < d_{j,21} + d_{j,22}$. Since d and θ is positive, hence $U_j < d_{j,11} + d_{j,12} + d_{j,21} + d_{j,22} < d_{j,11} + d_{j,21} + \theta (d_{j,12} + d_{j,22}) = d_j, \text{PIBI}$.

Therefore, when the algorithm always chooses individuals with small PIBI distance in the iteration process, the uncertainty will also decrease, and the upper bound of uncertainty will approach the PIBI distance infinitely. The PIBI indicator operation is executed as **Algorithm 4**.

PIBI indicator selects the individuals in the subspace. Firstly, step 3 calculates the PIBI value of each subpopulation in the subspace as **Eq. (9)**; step 4 is to sort the individuals in the subspace according to PIBI value; step 8 selects the individual ranked first by PIBI value each time, then puts it into S_t^{PIBI} , and sets the corresponding position INF in ranking; **Algorithm 4** stops when the number of individuals in the iterative guidance S_t^{PIBI} is N . PIBI indicator ensures that the population evolves in the direction of low uncertainty.

4. Experiment and Analysis

4.1. Benchmarks and evaluation metrics

In this paper, test suites IDTLZ, IMFG, and IMAF are extended from the typical MaOPs such as DTLZ, WFG and MaF, which are heavily used to test the performance of algorithms [31,45,46]. In addition, the IDTLZ1–4, IDTLZ7, IWFG1–9, IMAF1–6, IMAF8–9, IMAF13–14 are employed to evaluate the performance of the IMAOA-DRA as test suites. The reason for intercepting IMAF is that IMAF7, and IMAF10–12 are included in IDTLZ and IWFG, and the reason for intercepting IDTLZ is that IDTLZ5 and IDTLZ6 do not have well Pareto front when the number of objectives exceeds three [55]. The characteristics of the test functions are presented in **Table 2**.

The performance indicators are interval Hypervolume metric (IHV) [56], Interval inverse generational distance (IIGD) [14], and Imprecision (IM) [14], which are popular in interval optimization. IHV is the interval form of the hypervolume metric, which measures the combined performance of convergence and distribution. A larger value of IHV that the final solution set has better combined performance. IIGD is the interval form of inverse generational distance metric, which reflects the convergence more. A smaller IIGD that means the final solution set is closer to the true Pareto front. IM reflects the uncertainty of the solution set, which smaller IM illustrates the more precise and stable final solution set. In addition, the experimental parameters are set in **Table 3**.

Table 8

IM (mean) obtained across IDTLZ 1–4.

M	Pro.	DI-μMOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA	Pro.	DI-μMOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA
4	IDT	2.6624e-1(-)	9.3562e-2(=)	1.4927e+0(-)	1.2974e-1(-)	1.2246e-2	IDT	4.4591e-1(-)	1.8121e-2(-)	1.2496e+0(-)	6.4053e-2(+)	3.9686e-1
5	LZ1	1.1816e-1(-)	1.5468e-1(-)	4.6974e-1(-)	1.7004e-1(-)	3.3648e-3	LZ3	1.4811e-1(-)	1.6542e-1(-)	4.2286e-1(-)	1.9526e-1(-)	5.3637e-2
6		8.4277e-2(-)	7.1981e-2(-)	1.5637e-1(-)	6.6931e-2(-)	2.1533e-3		1.0524e-1(-)	8.3813e-2(=)	1.2778e-1(-)	9.0908e-2(=)	1.5602e-2
8		1.4249e-2(-)	1.2519e-2(-)	1.7553e-2(-)	7.4251e-3(-)	8.7994e-5		1.1297e-2(-)	1.1836e-2(-)	1.4615e-2(-)	1.0936e-2(-)	1.7518e-3
10		9.9494e-4(-)	1.1684e-3(-)	1.7613e-3(-)	1.1809e-3(-)	4.3660e-5		1.7082e-3(-)	1.4049e-3(-)	1.6532e-3(-)	1.6892e-3(-)	1.5712e-4
4	IDT	1.0332e-1(-)	1.1487e+0(-)	1.2526e+0(-)	4.4609e-1(-)	3.1060e-3	IDT	9.2659e-1(-)	4.4493e-1(-)	1.4078e+0(-)	5.3763e-1(-)	1.2234e-3
5	LZ2	3.0529e-1(-)	2.3401e-1(-)	4.4080e-1(-)	1.9134e-1(-)	2.1845e-3	LZ4	2.9935e-1(-)	2.3038e-1(-)	4.2973e-1(-)	1.7438e-1(-)	1.8233e-3
6		6.9793e-2(-)	1.0079e-1(-)	1.5709e-1(-)	7.4331e-2(-)	4.2671e-3		1.0339e-1(-)	9.4062e-2(-)	1.4026e-1(-)	8.9207e-2(-)	1.8889e-4
8		1.2607e-2(-)	9.5344e-3(-)	1.7831e-2(-)	1.0103e-2(-)	8.7560e-4		8.6730e-3(-)	1.2794e-2(-)	1.6890e-2(-)	1.0523e-2(-)	1.0100e-4
10		1.8058e-3(-)	1.4491e-3(-)	1.9092e-3(-)	1.3289e-3(-)	1.2160e-4		1.5520e-3(-)	1.5737e-3(-)	1.6618e-3(-)	1.3498e-3(-)	1.1373e-5
+/-/=		0/20/0		0/18/2			0/20/0		1/18/1			

Table 9The p value of statistical tests on IDTLZ.

Compared algorithm: IMaOEA-DRA			Test method	DI-μMOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD
IDTLZ	IIGD	p value	Friedman	0.0017	7.74e-06	3.06e-4	5.70 e-05
			Wilcoxon	0.0051	8.86e-05	2.19e-4	3.90e-04
	IHV	p value	Friedman	7.74e-06	7.74e-06	0.0017	7.74e-06
			Wilcoxon	8.85e-05	8.85e-05	8.92e-4	8.85e-05
	IM	p value	Friedman	7.74e-06	5.70e-05	7.74e-06	5.70e-05
			Wilcoxon	8.85e-05	0.0010	8.85e-05	0.0012

 $p < 0.05$ indicates rejection of the null hypothesis below the 0.05 significance level;

4.2. Time complexity analysis

This section analyses the time complexity of the IMaOEA-DRA, involving DIP-dominance strategy, clustering strategy, and PIBI indicator, and it is mainly in mating selection and environment selection. For a population of size N , mating selection (step 4, Algorithm 1) adopt the tournament selection whose individuals need to compared in pairs, and its worst-case time complexity is $O(N^2)$. DIP-dominance ranking (Step7, Algorithm 1) first sorts the individuals in ascending order of their objectives successively with heapsort (Step2, Algorithm 2), and its worst-time complexity is $O(N \log N)$.

Secondly (Step6–20, Algorithm 2), the sequential ordering is used in Pareto dominance ranking, and its time complexity of comparison between non-dominated front is $O(0)$ when all solutions are non-dominated solutions, that is, all are in the same front; When the latter front is dominated by the previous front, the time complexity of inter-front comparison is $O(N)$, and the time complexity of individual pairwise comparison in the front is $O(N^2)$; since each individual has M objectives, and each objective has upper and lower bounds, the worst-case time complexity of DIP-dominance is $O(2MN^2)$.

Algorithm 3 first performs clustering operation which calculates the distance between two individuals with lower objectives (Step3–10, Algorithm 3) and its time complexity of the algorithm is $O(N^2)$, and then it calculates the PIBI indicator for each individual, its time complexity is $O(2N)$ because of each individual needs to calculate the upper and lower bound values separately; hence, the worst-case time complexity of the PIBI indicator is $O(N^2)$. Taken together, the worst-case time complexity of the IMaOEA-DRA is $O(2MN^2)$.

4.3. Parameter sensitivity experiment

In this section, a parameter experiment was set up to demonstrate the degree of influence of parameter α in DIP-dominance on IMaOEA-DRA. The value of parameter α ranges from $[0, \beta]$, the closer to 0, the steeper the decay curve in late iteration and the gentler in the early iteration. Different uncertainty attenuation the degree will directly affect the convergence performance of IMaOEA-DRA, so we analyze the execution efficiency of the algorithm under different α values by IIGD and IM, taking IDTLZ2 as an example in Fig. 5. In the test, we designed $\alpha = \beta$

/step, and the step size was set at 100, 200, 300, 400, 500, 600, 700, 800, 1000.

The chosen value directly determines the rate of uncertainty decline but does not affect the final uncertainty value since the convergence of the decay function towards unity and remains constant at the point of convergence as described in the Eq. (7).

As depicted in Fig. 5, there exists a conflicting relationship between convergence and uncertainty where better uncertainty typically implies poorer convergence. This indicates that selecting an appropriate step size is a multi-objective problem characterized by conflicting objectives without an optimal solution in objective meaning. The numerical results indicate that the influence of the parameters on convergence and uncertainty is minimal.

4.4. Ablation experiment

In this section, the ablation experiment of the IMaOEA-DRA was set up, which consists of IMaOEA-DRA without DIP-dominance and PIBI indicator (IMaOEA, I for short), IMaOEA + DIP-dominance (IMaOEA+D, D for short), IMaOEA + PIBI indicator (IMaOEA+P, P for short) and proposed IMaOEA-DRA (DRA for short).

The IMaOEA is a standard decomposition-based algorithm, which chooses the upper bound value of the interval individual as the objective value; IMaOEA+D simply applies DIP-dominance, and the rest is the same as IMaOEA. IMaOEA+P simply applies PIBI indicator, and the rest is the same as IMaOEA. In this manner, the comparative analysis of diverse strategies in IMaOEA-DRA was conducted. Specifically, we evaluate the IIGD, IHV, and IM metrics across IDTLZ 1–4 and IDTLZ7 test suites as presented in Tables 4–6.

For IIGD, on the five test functions of IDTLZ, IMaOEA-DRA is all better than IMaOEA; on 4 objectives, IMaOEA-DRA is superior to IMaOEA+P and IMaOEA+D by three test functions and five test functions, respectively; on 8 objectives, IMaOEA-DRA is better than IMaOEA+P and IMaOEA+D by two and all test functions, respectively; on the 10 objectives, IMaOEA-DRA has four excellent test functions compared with IMaOEA+P and IMaOEA+D. For IHV, IMaOEA-DRA also has better performance, that is, three test functions on 4, 8, 10 objectives compared with IMaOEA+P and all functions compared with IMaOEA+D. For IM, IMaOEA-DRA only performs worse than

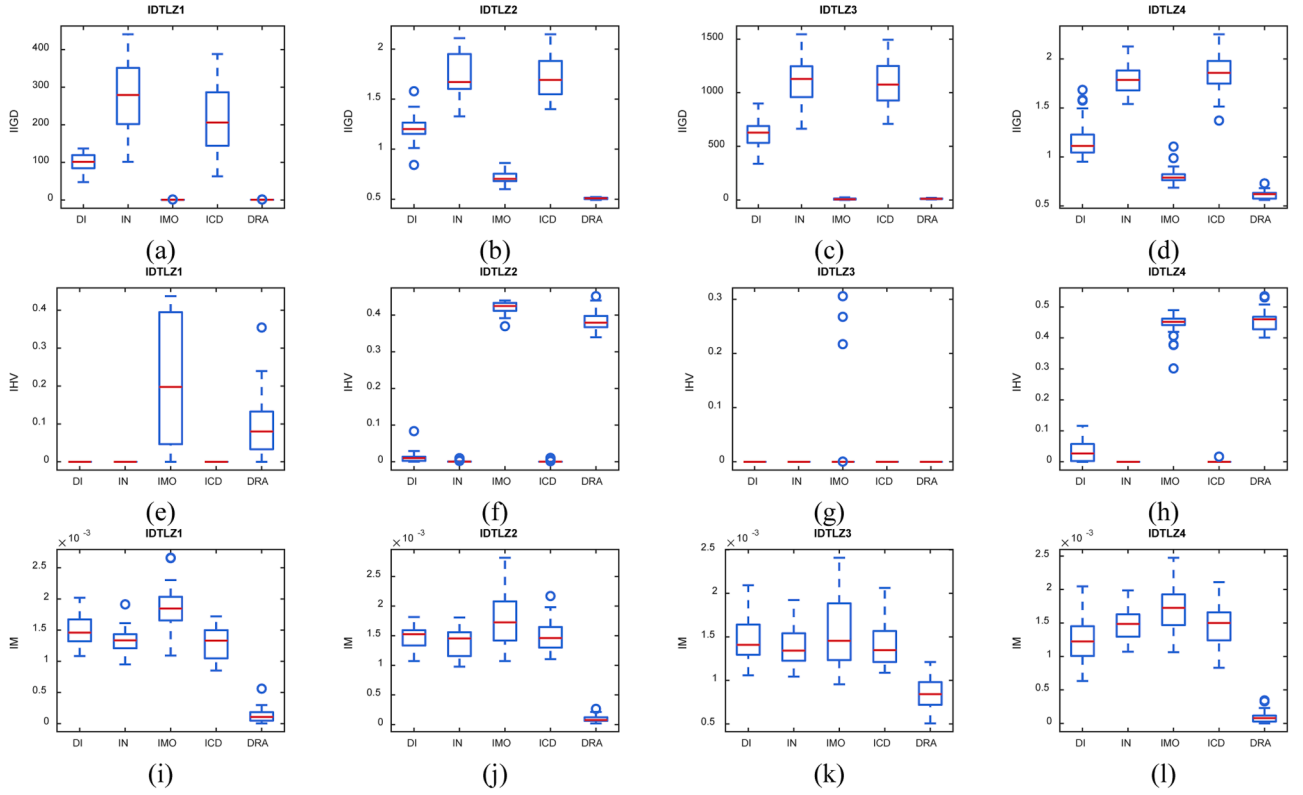


Fig. 7. Box-plot of IIGD, IHV, IM with 10 objectives on the IDTLZ1–4.

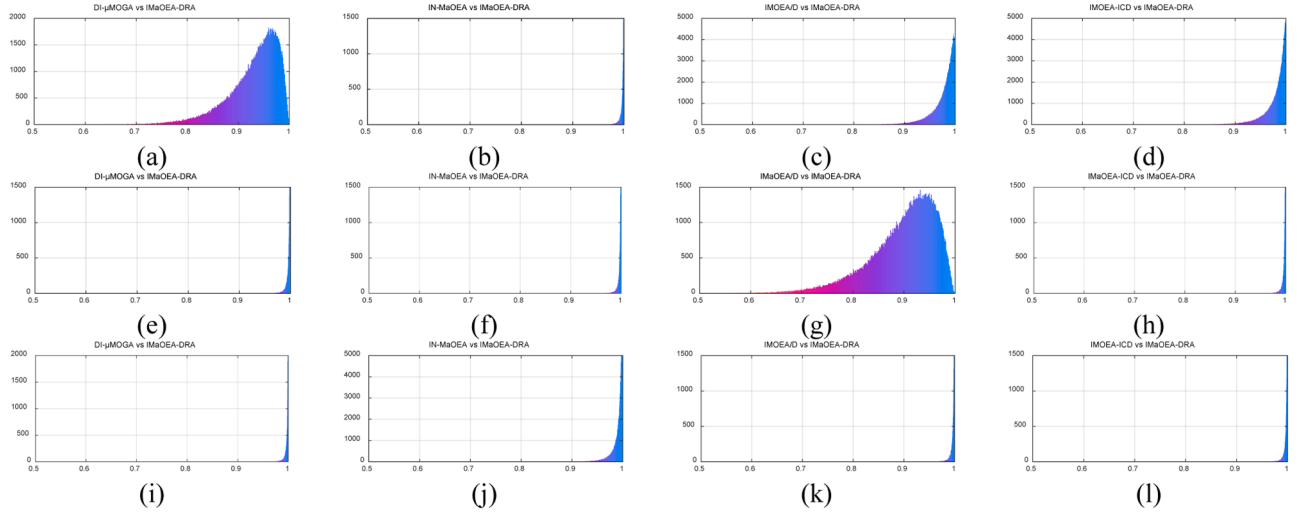


Fig. 8. Sample distribution across the four algorithms in the Bayesian test by IDTLZ test suites. (a)–(d): IIGD, (e)–(h): IHV, (i)–(l): IM.

Table 10

The Probability of dominance obtained across IDTLZ in Bayesian tests.

IMaOEA-DRA		DI	IN	IMO	ICD
IIGD	Left	0.000	0.000	0.000	0.000
	Rope	3.466	0.000	93.876	0.056
	Right	96.534	100.00	6.124	99.944
IHV	Left	0.000	0.000	0.008	0.000
	Rope	0.002	0.000	83.438	0.000
	Right	99.998	100.0	16.554	100.0
IM	Left	0.000	0.000	0.000	0.000
	Rope	0.000	0.000	0.000	0.000
	Right	100.0	100.0	100.0	100.0

IMaOEA+P on IDTLZ2 and IDTLZ3 of the 10 objectives, and worse than IMaOEA on IDTLZ7 of 8 objectives. It performs best on the rest of the test functions.

In summary, both the DIP-dominance and PIBI indicator proposed in this paper show good results on IMaOPs. IMaOEA+D and IMaOEA+P adding DIP-dominance and PIBI indicator separately, compared with IMaOEA on the three metrics, the performance of the final solution set is improved. Particularly, IMaOEA-DRA, which combines the two strategies, shows the best performance in the experiment.

We can see that for around 75% of the test problems, uncertainty, convergence and diversity are simultaneously excellent; while, IMaOEA is easy to fall into a local optimal solution and cannot solve the corresponding metrics with a single strategy, such as IMaOEA algorithm on

Table 11

IM (mean) obtained across IWFG 1–9.

M	Pro.	DI- μ MOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA	Pro.	DI- μ MOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA
4	IWF	1.1174e+0(-)	8.6769e-1(-)	1.1726e+0(-)	7.5215e-1(-)	1.3394e-1	IWF	1.1310e+0(-)	5.3225e-1(-)	1.1311e+0(-)	8.8084e-1(-)	8.7469e-2
5	G1	3.0632e-1(-)	3.0147e-1(-)	4.2660e-1(-)	3.1319e-1(-)	5.2819e-2	G6	3.8722e-1(-)	3.9577e-1(-)	3.9893e-1(-)	4.2458e-1(-)	4.6311e-2
6		1.0993e-1(-)	9.5065e-2(=)	1.3988e-1(-)	9.7903e-2(=)	1.6076e-2		1.3233e-1(-)	1.3120e-1(-)	1.5225e-1(-)	1.3071e-1(-)	1.5944e-2
8		1.5287e-2(-)	1.0619e-2(-)	1.6703e-2(-)	1.2420e-2(-)	1.8204e-3		1.2288e-2(-)	9.3693e-3(=)	1.6550e-2(-)	1.4636e-2(-)	2.7500e-3
10		1.1360e-3(-)	1.7851e-3(-)	1.6027e-3(-)	1.5093e-3(-)	9.2086e-5		1.1958e-3(-)	1.4561e-3(-)	1.7253e-3(-)	1.4630e-3(-)	1.2017e-4
4	IWF	9.0469e-1(-)	5.6597e-1(-)	1.2881e+0(-)	6.0901e-1(-)	2.0905e-2	IWF	9.6873e-1(-)	6.0247e-1(-)	1.2954e+0(-)	7.9287e-1(-)	5.2683e-2
5	G2	2.9579e-1(-)	2.6297e-1(-)	4.2189e-1(-)	2.1566e-1(-)	1.5225e-2	G7	3.6545e-1(-)	2.9706e-1(-)	3.8874e-1(-)	3.3826e-1(-)	2.8704e-2
6		1.0281e-1(-)	8.1525e-2(-)	1.4173e-1(-)	8.1227e-2(-)	8.8793e-3		1.1628e-1(-)	1.2222e-1(-)	1.4680e-1(-)	1.3517e-1(-)	1.1821e-2
8		1.3768e-2(-)	1.1346e-2(-)	1.5753e-2(-)	1.4792e-2(-)	1.5053e-3		1.5175e-2(-)	1.6627e-2(-)	1.5637e-2(-)	1.4548e-2(-)	2.0895e-3
10		1.4262e-3(-)	1.2138e-3(-)	1.7929e-3(-)	1.9312e-3(-)	3.5220e-5		1.0796e-3(-)	1.2713e-3(-)	1.6702e-3(-)	1.8360e-3(-)	9.1550e-5
4	IWF	1.1161e+0(-)	7.9977e-1(-)	1.3525e+0(-)	7.0717e-1(-)	1.2343e-1	IWF	1.0116e+0(-)	8.0670e-1(-)	1.2390e+0(-)	6.9474e-1(-)	2.3086e-1
5	G3	3.6705e-1(-)	3.8310e-1(-)	3.9741e-1(-)	4.2373e-1(-)	4.1679e-2	G8	3.6489e-1(-)	2.9074e-1(-)	3.9256e-1(-)	3.1235e-1(-)	7.5313e-2
6		1.2652e-1(-)	1.1227e-1(-)	1.3586e-1(-)	1.3556e-1(-)	1.4402e-2		1.2760e-1(-)	9.8399e-2(=)	1.1919e-1(-)	1.4035e-1(-)	2.3912e-2
8		1.5276e-2(-)	1.6731e-2(-)	1.5750e-2(-)	1.5256e-2(-)	9.3503e-4		1.3249e-2(-)	1.3269e-2(-)	1.3757e-2(-)	1.2650e-2(-)	2.8010e-3
10		1.9419e-3(-)	1.6317e-3(-)	1.5246e-3(-)	1.1643e-3(-)	1.1575e-4		1.3485e-3(-)	1.5584e-3(-)	1.6238e-3(-)	1.6880e-3(-)	1.9699e-4
4	IWF	1.1186e+0(-)	8.2513e-1(-)	1.2696e+0(-)	8.0913e-1(-)	1.0059e-1	IWF	1.1070e+0(-)	9.0152e-1(-)	1.2153e+0(-)	8.2017e-1(-)	1.1168e-1
5	G4	4.1902e-1(-)	3.7706e-1(-)	3.9554e-1(-)	3.3682e-1(-)	4.9720e-2	G9	3.8696e-1(-)	3.8463e-1(-)	4.1016e-1(-)	3.8577e-1(-)	6.5440e-2
6		1.0283e-1(-)	9.9538e-2(=)	1.3147e-1(-)	1.1896e-1(-)	1.8252e-2		1.0178e-1(-)	1.0909e-1(-)	1.2616e-1(-)	1.3187e-1(-)	2.9748e-2
8		1.8390e-2(-)	1.3766e-2(-)	1.5462e-2(-)	1.5952e-2(-)	1.2787e-3		1.2605e-2(-)	1.5900e-2(-)	1.4362e-2(-)	1.7855e-2(-)	3.5204e-3
10		1.6070e-3(-)	2.1897e-3(-)	1.7410e-3(-)	1.3107e-3(-)	9.4280e-5		1.6080e-3(-)	1.4231e-3(-)	1.6382e-3(-)	1.5385e-3(-)	2.6118e-4
4	IWF	1.2720e+0(-)	6.3367e-1(-)	1.2690e+0(-)	8.7335e-1(-)	4.1366e-2						
5	G5	3.9723e-1(-)	4.5341e-1(-)	4.3048e-1(-)	3.9543e-1(-)	2.3413e-2						
6		1.6375e-1(-)	1.2683e-1(-)	1.3658e-1(-)	1.3783e-1(-)	1.0652e-2						
8		1.2333e-2(-)	1.3949e-2(-)	1.5572e-2(-)	1.2795e-2(-)	1.7850e-3						
10		1.5166e-3(-)	1.1955e-3(-)	1.7650e-3(-)	1.5135e-3(-)	6.0501e-5						
+/-/=		0/45/0			0/41/4			0/45/0		0/44/1		

10-objective IDTLZ3.

Notably, for the disconnected interval test problem like IDTLZ7, all of the above algorithms do not solve it well, which the final solutions fall into the local optimum; therefore, the value of IDTLZ7 has no reference significance.

In addition, IMaOEA+D which absence PIBI indicator, the final solution on IDTLZ1 and IDTLZ3 appeared to have difficult convergence and poor distribution. Fig. 6 demonstrates that the IMaOEA-DRA exhibits superior convergence, diversity, and stability compared to the other three methods.

Even though individual metrics may not be as exceptional, the observed differences are negligible in magnitude. In particular, on uncertainty, the IMaOEA-DRA shows exceptional performance in addressing uncertainty without compromising convergence and diversity.

4.5. Contrast experiment

In this section, we present a comparative analysis of IMaOEA-DRA with three state-of-the-art interval evolutionary algorithms: DI- μ MOGA [20], IN-MOEA [21], IMOEA/D [57], IMOEA-ICD [58]. These algorithms are evaluated on three test suites: IDTLZ, IWFG and IMaF. DI- μ MOGA utilizes the interval possible degree dominance relation for its operation, while IMOEA-ICD incorporates the interval crowding distance in environment selection. On the other hand, IN-MOEA employs the interval credibility strategy during environment selection.

The values in the table are the mean values obtained when the algorithm runs 30 times on a dataset, and (+/-/=) are the values that perform better/worse/similar to our algorithm, respectively.

4.5.1. IDTLZ test suites

The performance metrics IIGD, IHV, and IM obtained across IDTLZ test suites are presented in Tables 7 and 8. On the IDTLZ test suite, IMaOEA-DRA shows superior performance. For 90%, 100%, 100% of test functions, IMaOEA-DRA outperforms DI- μ MOGA in terms of IIGD, IHV and IM metrics. Additionally, for 100%, 100%, 90% of test functions, IMaOEA-DRA is superior to IN-MaOEA on IIGD, IHV and IM metrics. For 90%, 85%, 100% of test functions, IMaOEA-DRA consists of better values than IMOEA/D. Furthermore, for 75%, 100%, 90% of test

functions, IMaOEA-DRA take over the lead IMOEA-ICD on IIGD, IHV and IM metrics.

Compared to the other four algorithms, our method exhibits significant enhancements in all three metrics. Specifically, IMaOEA-DRA enhances the IM metrics by around 97.5%, while preserving the inherent benefits of IIGD and IHV.

Table 9 presents the outcomes of the statistical tests conducted on the IDTLZ test suites, encompassing both Wilcoxon sign rank and Friedman rank sum tests. Based on the obtained p -values, it is evident that the experimental results achieved by our proposed algorithm exhibit significant disparities when compared to those attained by existing algorithms across all three indicators. Fig. 7 presents the statistical box plots of the five algorithms applied to 10 objectives, revealing a distinct advantage exhibited by IMaOEA-DRA.

Furthermore, considering that the focus of our research is to address uncertain optimization problems, we drew upon the Bayesian testing methods presented in the literature [59,60] to perform a detailed analysis of the performance of IMaOEA-DRA and the comparison algorithms across three evaluation metrics.

Specifically, the prior parameters of the Dirichlet distribution were set as $s = 0.5$ and $z_0 = 0$. We then sampled 150,000 points in the space using the Monte Carlo method.

In Fig. 8, we present the probability distribution of IMaOEA-DRA outperforming the comparison algorithms across three metrics without the introduction of Rope. Additionally, Table 10 shows the probability that the algorithm with Rope introduced outperforms the comparison algorithms. In the table, 'Left' represents the current comparison algorithm, while 'Right' represents our proposed algorithm.

By incorporating prior information, the Bayesian method can still provide reliable statistical inference even in cases of high data uncertainty or small sample sizes. From Fig. 8 and Table 10, we can observe that the significance differences identified by the Bayesian test are largely consistent with the results obtained using the Friedman and Wilcoxon tests. This further underscores the robustness and adaptability of our proposed algorithm.

4.5.2. IWFG test suites

Based on the analysis presented in Tables 11 and 12, our algorithm demonstrates superior performance in terms of the IWFG test suite.

Table 12

IIGD (mean) and IHV (mean) obtained across IWFG1–9.

Metric		IIGD					IHV				
Pro.	M	DI- μ MOGA	IN-MaOEA	IMOE/D	IMOE/ICD	IMaOEA-DRA	DI- μ MOGA	IN-MaOEA	IMOE/D	IMOE/ICD	IMaOEA-DRA
IWF	4	1.6060e+0(-)	9.7025e-1(-)	7.9197e-1(-)	1.9146e+0(-)	5.3826e-1	4.6178e-1(-)	7.6197e-1(-)	7.7449e-1(-)	3.0460e-1(-)	8.2836e-1
	5	1.5418e+0(-)	1.3836e+0(-)	1.0927e+0(-)	1.3213e+0(-)	6.8273e-1	6.7028e-1(-)	7.8272e-1(=)	7.8270e-1(=)	7.4835e-1(-)	8.0595e-1
	6	2.2470e+0(-)	1.6139e+0(-)	1.2944e+0(-)	1.5015e+0(-)	8.5315e-1	6.9131e-1(-)	7.2950e-1(-)	7.8450e-1(=)	7.6746e-1(-)	8.1245e-1
	8	2.8363e+0(-)	2.0435e+0(-)	1.7944e+0(-)	2.0666e+0(-)	1.4637e+0	3.7853e-1(-)	6.9858e-1(-)	8.2975e-1(=)	5.1166e-1(-)	8.3728e-1
G1	10	2.6742e+0(-)	2.1466e+0(-)	1.9549e+0(-)	2.6125e+0(-)	1.7804e+0	4.1611e-1(-)	6.1558e-1(-)	8.2642e-1(=)	4.6564e-1(-)	8.5071e-1
	4	7.7450e-1(-)	1.2160e+0(-)	6.1953e-1(-)	1.3510e+0(-)	3.9555e-1	9.0843e-1(=)	8.7704e-1(=)	9.0131e-1(=)	8.6708e-1(-)	9.1405e-1
	5	1.3486e+0(-)	1.4781e+0(-)	8.9109e-1(-)	1.7816e+0(-)	6.0204e-1	9.1620e-1(=)	8.9437e-1(=)	9.1766e-1(=)	8.8058e-1(-)	9.3293e-1
	6	1.4914e+0(-)	3.4210e+0(-)	1.0475e+0(-)	3.0372e+0(-)	7.9240e-1	9.3843e-1(=)	7.5308e-1(-)	9.1916e-1(=)	7.8864e-1(-)	9.2775e-1
G2	8	1.9950e+0(=)	4.8546e+0(-)	1.7600e+0(-)	7.1544e+0(-)	1.4454e+0	8.8619e-1(=)	7.6065e-1(-)	9.2801e-1(=)	5.8869e-1(-)	9.2191e-1
	10	2.0243e+0(=)	9.9027e+0(-)	1.9062e+0(=)	8.8246e+0(-)	1.8683e+0	9.0666e-1(=)	5.8481e-1(-)	9.2833e-1(=)	7.0156e-1(-)	9.3101e-1
	4	7.9493e-1(-)	1.0939e+0(-)	3.4514e-1(+)	1.2827e+0(-)	3.8602e-1	9.3057e-2(-)	6.1208e-2(-)	2.2719e-1(+)	8.0858e-2(-)	1.8497e-1
	5	1.0091e+0(=)	2.2040e+0(-)	6.5456e-1(+)	2.5626e+0(-)	1.0659e+0	5.6762e-2(+)	1.0918e-2(-)	1.3441e-1(+)	7.9165e-3(-)	4.8812e-2
G3	6	2.0776e+0(-)	2.9765e+0(-)	1.1231e+0(+)	3.2992e+0(-)	1.4913e+0	7.7100e-2(+)	0.000e+0(-)	7.7273e-2(+)	1.1074e-2(-)	2.2342e-2
	8	2.3628e+0(+)	4.9822e+0(-)	2.3050e+0(+)	4.5120e+0(-)	3.8887e+0	7.2750e-2(+)	0.000e+0(-)	9.5243e-3(+)	0.000e+0(-)	2.7230e-3
	10	3.2539e+0(+)	6.3296e+0(-)	3.9606e+0(+)	5.4588e+0(-)	5.8548e+0	0.000e+0(=)	0.000e+0(=)	0.000e+0(=)	0.000e+0(=)	0.000e+0
	4	1.0357e+0(-)	1.4434e+0(-)	1.1857e+0(-)	1.4760e+0(-)	8.2761e-1	5.0231e-1(-)	4.9824e-1(-)	5.1173e-1(-)	4.8479e-1(-)	5.7821e-1
IWF	5	1.6596e+0(+)	2.2954e+0(-)	1.9365e+0(-)	2.0705e+0(-)	1.8041e+0	5.2852e-1(-)	5.2801e-1(-)	5.5391e-1(=)	5.4158e-1(=)	5.6310e-1
	6	2.1672e+0(+)	3.2542e+0(=)	2.9313e+0(+)	3.0848e+0(+)	3.2639e+0	5.4999e-1(+)	4.9986e-1(-)	5.8606e-1(+)	5.1236e-1(-)	5.1832e-1
	8	3.9824e+0(+)	5.5725e+0(-)	4.9043e+0(-)	4.9513e+0(-)	4.5821e+0	4.3548e-1(-)	4.5408e-1(-)	5.8805e-1(=)	5.0659e-1(-)	5.8201e-1
	10	6.6300e+0(+)	7.6991e+0(+)	7.4326e+0(+)	7.7841e+0(=)	7.7917e+0	4.2135e-1(-)	4.3473e-1(-)	5.7115e-1(+)	4.5686e-1(-)	5.3438e-1
G4	4	9.7210e-1(-)	1.4176e+0(-)	1.1533e+0(-)	1.4812e+0(-)	8.2049e-1	4.8593e-1(-)	4.9246e-1(-)	4.8634e-1(-)	4.4582e-1(-)	5.5476e-1
	5	1.6614e+0(=)	2.1643e+0(-)	1.9400e+0(-)	2.1044e+0(-)	1.7398e+0	4.3807e-1(-)	5.0303e-1(-)	5.2835e-1(=)	5.2026e-1(-)	5.4933e-1
	6	2.2292e+0(+)	3.1065e+0(+)	3.0271e+0(+)	3.5677e+0(-)	3.1845e+0	4.8202e-1(-)	4.7290e-1(-)	5.5323e-1(+)	4.7911e-1(-)	5.1321e-1
	8	4.5383e+0(-)	5.2477e+0(-)	4.6119e+0(-)	5.2356e+0(-)	4.2669e+0	3.3601e-1(-)	4.2413e-1(-)	6.1656e-1(+)	4.6175e-1(-)	5.9070e-1
G5	10	5.7814e+0(+)	7.6968e+0(=)	7.0471e+0(+)	6.6896e+0(+)	7.7405e+0	3.6065e-1(-)	4.3282e-1(-)	5.8641e-1(+)	4.8683e-1(-)	5.2279e-1
	4	1.1908e+0(-)	1.8643e+0(-)	1.1945e+0(-)	1.7400e+0(-)	8.2622e-1	4.9526e-1(-)	4.2023e-1(-)	4.8017e-1(-)	4.2439e-1(-)	5.4608e-1
	5	1.7909e+0(-)	2.7711e+0(-)	2.0617e+0(-)	2.2081e+0(-)	1.7058e+0	4.9836e-1(-)	4.4964e-1(-)	4.8292e-1(-)	4.9701e-1(-)	5.4606e-1
	6	2.7588e+0(+)	3.2979e+0(-)	3.3006e+0(-)	3.4543e+0(-)	3.1267e+0	4.3046e-1(-)	4.6971e-1(-)	5.0570e-1(-)	5.3243e-1(+)	4.9051e-1
G6	8	4.6125e+0(+)	5.7242e+0(-)	5.3937e+0(-)	5.4038e+0(-)	4.8553e+0	3.9735e-1(-)	4.5208e-1(=)	5.1457e-1(+)	4.3559e-1(=)	4.4793e-1
	10	6.1674e+0(+)	7.3592e+0(+)	9.0578e+0(-)	7.0763e+0(+)	8.0436e+0	4.2353e-1(+)	4.9242e-1(+)	4.5501e-1(+)	4.2614e-1(+)	4.0047e-1
	4	1.0404e+0(-)	1.5104e+0(-)	1.1789e+0(-)	2.2351e+0(-)	8.1991e-1	4.9987e-1(-)	4.9212e-1(-)	5.2342e-1(-)	4.6470e-1(-)	5.8160e-1
	5	1.6371e+0(+)	2.7282e+0(-)	2.0077e+0(-)	2.3254e+0(-)	1.7667e+0	5.5704e-1(=)	5.1802e-1(-)	5.4846e-1(-)	5.5138e-1(=)	5.7159e-1
G7	6	2.5135e+0(+)	2.9297e+0(-)	3.3520e+0(-)	3.1530e+0(=)	3.1245e+0	4.3850e-1(-)	5.3686e-1(=)	5.4938e-1(+)	5.7364e-1(+)	5.1946e-1
	8	4.6650e+0(-)	5.3558e+0(-)	4.9945e+0(-)	4.5167e+0(-)	4.2769e+0	4.0478e-1(-)	4.6134e-1(-)	6.0840e-1(+)	5.0453e-1(=)	5.0999e-1
	10	6.6410e+0(+)	7.8880e+0(-)	6.9897e+0(-)	7.4482e+0(+)	7.7289e+0	4.0714e-1(-)	3.3966e-1(-)	6.3526e-1(+)	5.1122e-1(=)	5.1251e-1
	4	1.0516e+0(-)	1.4691e+0(-)	1.2373e+0(-)	1.8746e+0(-)	8.3748e-1	4.1176e-1(-)	3.7378e-1(-)	4.1856e-1(-)	3.2785e-1(-)	5.2029e-1
IWF	5	2.0390e+0(-)	2.5274e+0(-)	2.0318e+0(-)	2.5076e+0(-)	1.5810e+0	3.5814e-1(-)	3.7934e-1(-)	4.0992e-1(-)	3.4823e-1(-)	5.3512e-1
	6	2.7515e+0(=)	3.4887e+0(-)	3.1557e+0(-)	3.2794e+0(-)	2.8317e+0	4.0172e-1(-)	4.1726e-1(-)	3.8277e-1(-)	4.2855e-1(-)	5.0732e-1
	8	4.7327e+0(-)	5.9050e+0(-)	4.3501e+0(+)	5.0124e+0(-)	4.4946e+0	2.9007e-1(-)	3.6846e-1(-)	5.3583e-1(=)	4.1641e-1(-)	4.9736e-1
	10	6.7990e+0(+)	7.9434e+0(-)	6.8511e+0(+)	7.3120e+0(+)	7.4699e+0	1.8825e-1(-)	3.6247e-1(=)	5.0837e-1(+)	4.2835e-1(+)	3.7454e-1
G8	4	8.9884e-1(-)	1.5069e+0(-)	1.1207e+0(-)	1.3272e+0(-)	8.1127e-1	4.9493e-1(-)	4.6485e-1(-)	4.7749e-1(-)	4.5349e-1(-)	5.6418e-1
	5	1.6683e+0(=)	1.8576e+0(-)	1.9311e+0(-)	1.9123e+0(-)	1.6864e+0	4.7154e-1(-)	5.0436e-1(-)	5.0417e-1(-)	5.3231e-1(=)	5.5121e-1
	6	2.5709e+0(+)	2.6425e+0(+)	2.8751e+0(+)	3.0217e+0(-)	2.9522e+0	3.5567e-1(-)	4.8262e-1(-)	5.3628e-1(=)	4.5122e-1(-)	5.1932e-1
	8	4.4669e+0(-)	5.1523e+0(-)	4.6067e+0(-)	4.6631e+0(-)	4.1746e+0	3.0006e-1(-)	4.4813e-1(-)	5.4772e-1(=)	4.3877e-1(-)	5.6476e-1
G9	10	6.3289e+0(+)	7.6300e+0(-)	6.6754e+0(+)	7.0127e+0(-)	7.5582e+0	2.6849e-1(-)	3.3056e-1(-)	5.3666e-1(+)	5.0210e-1(+)	4.7449e-1
	+	17/22/6	4/39/2	13/31/1	5/38/2		5/33/7	1/37/7	16/12/17	4/34/7	

However, it is worth noting that in comparison with the DI- μ MOGA, the IGD metric shows no significant advantage, with only 42.2% of the test functions in IMaOEA-DRA performing better than DI- μ MOGA. On the other hand, both IHV and IM metrics demonstrate superior performance compared to DI- μ MOGA. This reason can be attributed to the tendency of IMaOEA-DRA to prioritize minimizing uncertainty during iteration. Consequently, for deceptive and mixed problem as IWFG5 and IWFG9, IMaOEA-DRA is less effective than DI- μ MOGA which employee the interval possible degree dominance relation in discovering the optimal PF.

Similarly, compared to the IMOE/D, our proposed algorithm does not show a significant advantage in the IHV metric but demonstrates clear advantages in both the IIGD and IM metrics.

In addition, our algorithm consistently outperforms the comparison algorithms in terms of the IM metric across all problems. This highlights the effectiveness of our method and its potential for practical applications with uncertainty characteristics.

The characteristics of IWFG1 include being multimodal, disconnected, multiscale, nonseparable, and convex. Similarly, IWFG2 is multimodal, disconnected, multiscale, nonseparable, and convex. For

such problems, our algorithm performs best on the IWFG test suite in terms of IHV, IIGD, and IM.

For a degenerate problem like IWFG3, we observe that almost all algorithms do not perform well. The numerical results reflect that the solutions obtained by the five algorithms, including ours, exhibit large errors. Addressing the problem of degeneration is a crucial aspect that we will focus on in our future research. The statistical evaluation of the algorithms is presented in Table 13, encompassing the Wilcoxon sign rank and Friedman rank sum tests.

Based on the p -value analysis, IMaOEA-DRA exhibits significant distinctions from the other four algorithms concerning IHV and IM. Additionally, our algorithm demonstrates comparable performance to DI- μ MOGA regarding IIGD and IMOE/D regarding IHV. Fig. 9 depict the statistical outcomes of 30 experiments conducted on 10 objectives within the IWFG test suite. Based on the findings, although our algorithm has no obvious advantage in IIGD compared with DI- μ MOGA and no obvious advantage in IHV compared with IMOE/D, the difference between them is minimal. Conversely, IMaOEA-DRA demonstrates a significant advantage in relation to IHV and IM, outperforming the four

Table 13The p value of statistical tests on IWFG.

Compared algorithm: IMaOEA-DRA			Test method	DI- μ MOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD
IWFG	IIGD	p value	Friedman	0.6547	3.82e-06	1.13e-2	1.53e-05
			Wilcoxon	0.9595	1.57e-07	1.28e-2	8.92e-06
	IHV	p value	Friedman	1.40e-6	1.01e-08	0.7630	2.96e-4
			Wilcoxon	2.65e-7	6.54e-08	0.9907	1.27e-6
	IM	p value	Friedman	1.97e-11	1.97e-11	1.97e-11	1.97e-11
			Wilcoxon	5.18e-09	5.18e-09	5.18e-9	5.18e-09

 $p < 0.05$ indicates rejection of the null hypothesis below the 0.05 significance level;

comparison algorithms in most test functions. This aligns with our corollary.

Table 14 provides a detailed comparison of the Bayesian test results across different algorithms, presenting the dominance probability of each algorithm on multiple metrics. The Bayesian test, by calculating the dominance probabilities across the IIGD, IHV, and IM performance indicators, further validates the significant advantages of the proposed algorithm in many-objective optimization. Specifically, the Bayesian test results indicate that the proposed algorithm outperforms IN-MaOEA in terms of IIGD, and performs comparably to DI- μ MOGA, IMOEA/D, and IMOEA-ICD. In terms of IHV, its performance surpasses that of DI- μ MOGA and IMOEA-ICD, while being similar to that of IN-MaOEA and IMOEA/D. For IM, the proposed algorithm consistently outperforms all comparison algorithms, demonstrating its superiority in handling uncertainty.

Additionally, Fig. 10 displays the sample distribution plots for pairwise comparisons of each algorithm, offering a visual representation of the relative performance of the algorithms on the aforementioned metrics. From these plots, although the proposed algorithm performs similarly to the comparison algorithms on certain problems, it

outperforms them in the majority of test cases, especially in handling problem uncertainty.

Consistent with the results of the Friedman and Wilcoxon tests, the Bayesian test also confirms that the proposed algorithm dominates the IIGD, IHV, and IM metrics, further corroborating the effectiveness and superiority of the proposed approach.

4.5.3. IMAF test suites

On the IIGD metric, IMaOEA-DRA demonstrates superior and similar performance compared to DI- μ MOGA, IN-MaOEA, IMOEA/D, and IMOEA-ICD in 86%, 95%, 82%, and 84% of the test functions, respectively.

When considering the IHV metric, IMaOEA-DRA surpasses IN-MaOEA, DI- μ MOGA, and IMOEA-ICD for over 66% of the test functions. Furthermore, on the IM metric, there is a significant advantage observed in approximately 95% more test functions when comparing with other algorithms.

Table 15 displays the results of the significance analysis conducted using both the Wilcoxon sign rank and Friedman rank sum tests. Table 16 presents the dominance probability values from the Bayesian

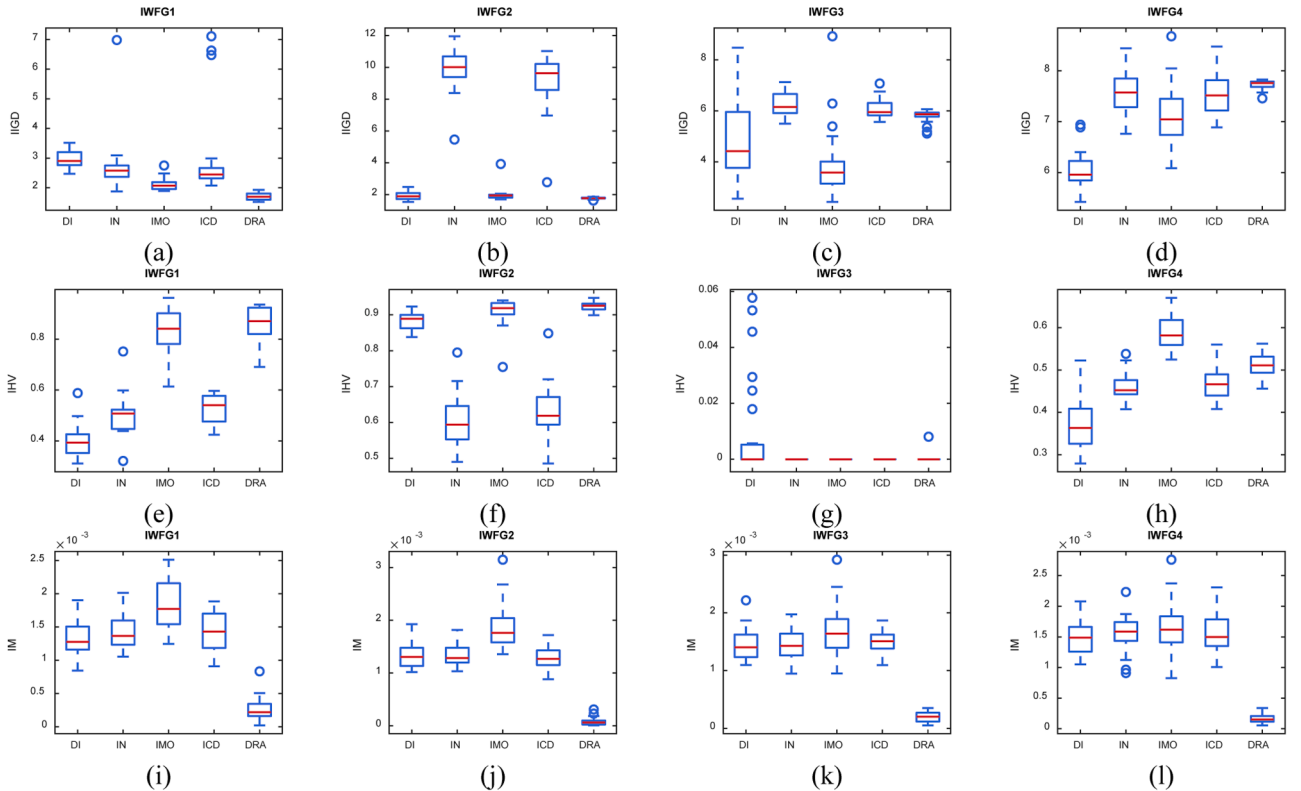
**Fig. 9.** Box-plot of IIGD, IHV, IM with 10 objectives on the IWFG1–4.

Table 14

The Probability of dominance obtained across IWFG in Bayesian tests.

IMaOEA-DRA		DI	IN	IMO	ICD
IIGD	Left	19.780	0.000	0.074	0.000
	Rope	77.238	3.734	99.896	98.326
	Right	2.982	96.266	0.030	1.674
IHV	Left	0.036	0.000	5.142	0.000
	Rope	16.936	99.986	94.340	1.916
	Right	83.028	0.014	0.518	98.084
IM	Left	0.000	0.000	0.000	0.000
	Rope	0.000	0.000	0.000	0.000
	Right	100.0	100.0	100.0	100.0

test on IMAF. Similarly, the settings for the Bayesian test are the same as those in Section 4.5.1.

Based on the obtained p -value, it is evident that IMaOEA-DRA exhibits statistically significant disparities in the three metrics of IMAF test suites when compared to the remaining four algorithms, except for IHV metric in IMOEA/D. The results of the Bayesian test are largely consistent with the Wilcoxon signed-rank and Friedman rank-sum tests, but the Bayesian test provides a more detailed probability distribution of dominance.

Tables 17 and 18 show the contrast experiments of the four algorithms on the IMAF test suites. While it is important to note that spatial characteristics may be lost in some test functions, it is evident that other algorithms do not outperform IMaOEA-DRA under the same conditions. The IHV of IMaOEA-DRA and IMOEA/D on the 6, 8, and 10 objectives in the IMAF1 and IMAF8 test suites is suboptimal.

However, our algorithm demonstrates significantly superior performance in handling uncertainty and convergence compared to IMOEA/D. Specifically, IMOEA/D surpasses our algorithm in IGD for only 8 out of

45 test functions, in IHV for 18 out of 45 test functions, and in IM for none of the 45 test functions.

Regarding the DI- μ MOGA, it outperforms our algorithm in IGD, IHV, and IM for only 6, 9, and 0 out of 45 test functions, respectively. Similarly, for IN-MaOEA and IMOEA-ICD, our algorithm consistently demonstrates superior performance across all three metrics, affirming its robustness and efficacy.

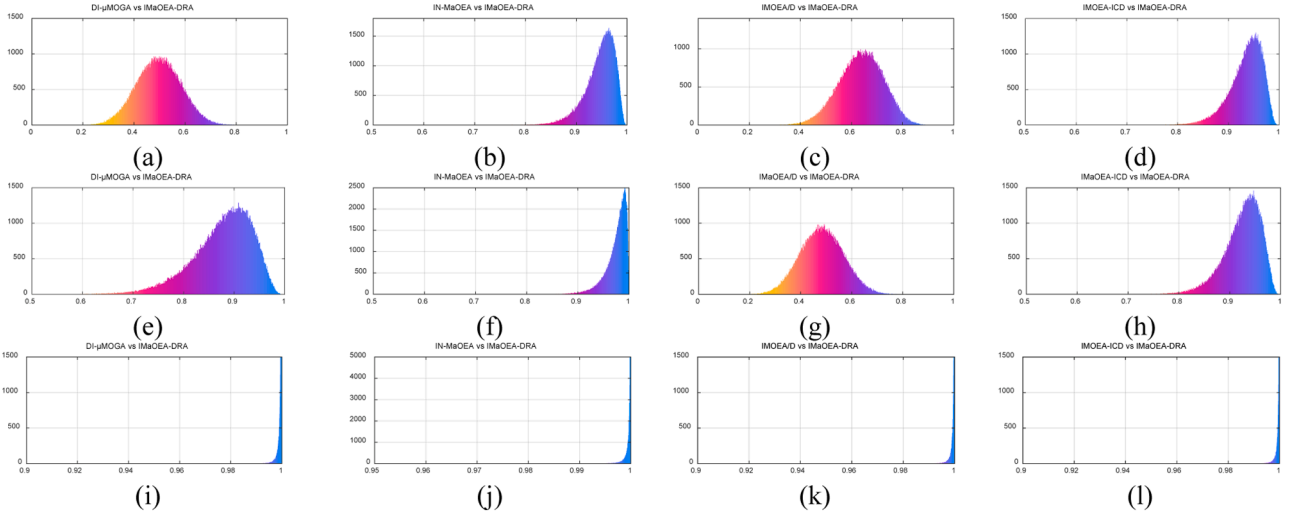
It is worth noting that the "0.000E +0" for IHV in the table indicates that the algorithm failed to obtain a valid solution set with spatial characteristics in multiple tests. While it is known that evolutionary algorithms possess inherent randomness, increasing the number of experiments could potentially address this issue. However, this also suggests that the algorithm's robustness may be suboptimal.

Fig. 11 depict the statistical findings obtained from conducting 30 experiments on a total of 10 objectives, and the statistical findings further substantiate the performance of IMaOEA-DRA. We observe that while the IMaOEA-DRA underperforms in IHV compared to the IMOEA/

Table 16

The Probability of dominance obtained across IMAF in Bayesian tests.

IMaOEA-DRA		DI	IN	IMO	ICD
IIGD	Left	0.000	0.000	0.000	0.000
	Rope	0.400	0.614	64.370	0.862
	Right	99.600	99.836	35.630	99.138
IHV	Left	0.000	0.000	0.000	0.000
	Rope	1.896	2.392	89.714	2.570
	Right	98.104	97.608	10.286	97.460
IM	Left	0.000	0.000	0.000	0.000
	Rope	0.000	0.000	0.000	0.000
	Right	100.0	100.0	100.0	100.0

**Fig. 10.** Sample distribution across the four algorithms in the Bayesian test by IWFG test suites. (a)-(d): IIGD, (e)-(h): IHV, (i)-(l): IM.**Table 15**The p value of statistical tests on IMAF.

Compared algorithm: IMaOEA-DRA		Test method		DI- μ MOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD
IMaF	IIGD	p value	Friedman	1.94e-04	3.47e-08	6.07e-4	1.54e-05
			Wilcoxon	3.88e-05	2.72e-07	1.75e-4	1.11e-06
	IHV	p value	Friedman	2.31e-2	2.47e-05	0.5371	9.89e-05
			Wilcoxon	1.64e-2	1.15e-05	0.7785	7.30e-05
IM	p value		Friedman	1.97e-11	3.48e-08	1.97e-11	3.48e-08
			Wilcoxon	5.18e-09	4.14e-07	5.18e-9	6.25e-07

$p < 0.05$ indicates rejection of the null hypothesis below the 0.05 significance level;

Table 17

IGD (mean) and IHV (mean) obtained across IMAF1–6, 8, 14, 15.

Metric	Pro.	M	IGD					IHV				
			DI- μ MOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA	DI- μ MOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA
F1	IMA	4	1.7722e-1(=)	2.7035e-1(-)	1.8631e-1(=)	3.5260e-1(-)	1.9682e-1	6.3156e-2(+)	4.7268e-2(+)	7.0376e-2(+)	4.2719e-2(+)	3.7557e-2
		5	3.0781e-1(-)	3.7882e-1(-)	2.6327e-1(=)	4.5323e-1(-)	2.3434e-1	1.5279e-2(+)	1.2283e-2(-)	2.3305e-2(+)	1.3211e-2(=)	1.3913e-2
		6	3.6682e-1(-)	4.7028e-1(-)	3.3398e-1(=)	5.1690e-1(-)	2.8818e-1	3.9320e-3(+)	3.1254e-3(+)	7.3089e-3(+)	2.7736e-3(+)	2.0002e-3
		8	4.4048e-1(-)	4.5715e-1(-)	4.2143e-1(-)	5.6549e-1(-)	3.1902e-1	9.0168e-5(-)	3.5195e-4(-)	6.3658e-4(+)	1.4818e-4(-)	4.6385e-4
		10	4.1741e-1(=)	4.2204e-1(-)	5.3618e-1(-)	6.1577e-1(-)	3.7940e-1	7.6346e-6(+)	1.5336e-5(+)	3.4947e-5(+)	5.8106e-6(-)	6.5720e-6
F2	IMA	4	1.1395e-1(=)	1.9878e-1(-)	1.4868e-1(=)	2.1545e-1(-)	1.2717e-1	3.0319e-1(=)	2.8016e-1(=)	2.8539e-1(=)	2.8605e-1(=)	2.9915e-1
		5	2.3441e-1(-)	2.6062e-1(-)	1.7941e-1(=)	2.7292e-1(-)	1.6312e-1	2.3362e-1(-)	1.9730e-1(-)	2.4792e-1(=)	2.0561e-1(-)	2.5729e-1
		6	2.1862e-1(=)	1.9640e-1(=)	2.4152e-1(-)	2.0857e-1(=)	1.6827e-1	2.5282e-1(=)	2.3032e-1(=)	2.5973e-1(-)	2.2996e-1(-)	2.4281e-1
		8	2.1773e-1(-)	2.2158e-1(=)	3.2461e-1(-)	2.3399e-1(=)	1.7605e-1	2.0921e-1(=)	1.8253e-1(=)	2.3797e-1(+)	1.9588e-1(=)	1.9383e-1
		10	2.5608e-1(=)	2.6421e-1(=)	3.8331e-1(-)	2.5628e-1(=)	2.4856e-1	2.0117e-1(+)	1.5979e-1(+)	2.1458e-1(+)	1.5260e-1(+)	1.2870e-1
F3	IMA	4	5.4500e+3(-)	4.0602e+4(-)	4.0414e+1(-)	6.5245e+5(-)	1.0830e+1	0.000e+0(-)	0.000e+0(-)	0.000e+0(-)	0.000e+0(-)	8.7732e-2
		5	6.3540e+4(-)	2.3143e+6(-)	5.2135e+1(-)	2.2510e+6(-)	2.7473e+1	0.000e+0(-)	0.000e+0(-)	8.3644e-2(-)	0.000e+0(-)	9.1527e-1
		6	2.7604e+5(-)	6.7079e+6(-)	2.3905e+1(-)	3.4812e+9(-)	2.9007e+1	0.000e+0(-)	0.000e+0(-)	7.3592e-2(=)	0.000e+0(-)	7.3784e-2
		8	6.5968e+5(-)	4.2284e+6(-)	8.7606e+1(-)	4.5993e+6(-)	2.0460e+1	0.000e+0(-)	0.000e+0(-)	0.000e+0(-)	0.000e+0(-)	1.4552e-2
		10	1.4904e+6(-)	4.3876e+6(-)	4.5993e+1(-)	4.5971e+6(-)	4.1334e+1	0.000e+0(-)	0.000e+0(-)	0.000e+0(-)	0.000e+0(-)	6.5656e-3
F4	IMA	4	7.0183e+1(-)	8.2077e+1(-)	6.8233e+0(-)	1.1154e+2(-)	2.4477e+0	0.000e+0(-)	0.000e+0(-)	1.7001e-2(-)	0.000e+0(-)	1.6815e-1
		5	1.4239e+2(-)	6.0758e+2(-)	1.5683e+1(-)	2.2474e+2(-)	8.6680e+0	0.000e+0(-)	0.000e+0(-)	1.9408e-3(-)	0.000e+0(-)	2.4244e-2
		6	7.1652e+2(-)	6.4172e+2(-)	3.1187e+1(-)	5.0158e+2(-)	2.4259e+1	0.000e+0(-)	0.000e+0(-)	7.3313e-5(-)	0.000e+0(-)	2.3227e-3
		8	2.8498e+3(-)	2.7175e+3(-)	1.8220e+2(-)	4.3755e+2(-)	1.0393e+2	0.000e+0(-)	0.000e+0(-)	6.1758e-7(-)	0.000e+0(-)	2.9047e-5
		10	9.9243e+3(-)	1.8670e+4(-)	6.5295e+2(-)	2.6873e+3(-)	5.3453e+2	0.000e+0(-)	0.000e+0(-)	1.1339e-8(-)	0.000e+0(-)	1.8397e-7
F5	IMA	4	1.4575e+0(=)	3.7616e+0(-)	2.8653e+0(-)	2.1702e+0(=)	1.8039e+0	5.5821e-1(+)	3.8519e-1(-)	4.5880e-1(-)	4.8780e-1(-)	5.0490e-1
		5	4.9593e+0(+)	5.0518e+0(=)	5.5586e+0(+)	5.5015e+0(+)	6.6798e+0	5.3232e-1(+)	4.4028e-1(=)	4.9466e-1(+)	3.3812e-1(-)	4.6290e-1
		6	9.2479e+0(+)	1.2705e+1(=)	1.3864e+1(+)	1.5849e+1(=)	1.4070e+1	5.5424e-1(+)	4.1130e-1(=)	4.0796e-1(=)	2.3749e-1(-)	4.1801e-1
		8	3.4723e+1(+)	7.5337e+1(-)	6.0738e+1(-)	5.8974e+1(=)	5.8111e+1	1.6279e-1(-)	0.000e+0(-)	4.5639e-1(=)	0.000e+0(-)	4.3369e-1
		10	1.1559e+2(+)	3.0648e+2(-)	2.5202e+2(-)	3.0045e+2(-)	2.2538e+2	0.000e+0(-)	0.000e+0(-)	4.2912e-1(=)	0.000e+0(-)	4.6723e-1
F6	IMA	4	5.0118e-2(=)	1.7267e-1(-)	4.9608e-2(=)	3.1432e-2(+)	5.2147e-2	1.3450e-1(-)	6.7389e-2(-)	1.7877e-1(-)	1.7197e-1(-)	1.9050e-1
		5	5.7050e-2(-)	4.2437e-1(-)	4.1590e-2(+)	4.5388e-2(=)	4.4491e-2	1.2927e-1(=)	4.8572e-2(-)	1.4639e-1(+)	1.1759e-1(-)	1.3774e-1
		6	3.0607e-1(-)	3.1325e-1(-)	1.1983e-1(=)	6.0881e-2(+)	1.1903e-1	8.7477e-2(+)	7.4862e-2(+)	1.3577e-1(+)	7.4838e-2(+)	6.7511e-2
		8	8.7284e-1(-)	5.8914e-1(-)	1.3370e-1(=)	6.6280e-2(+)	1.2636e-1	0.000e+0(-)	1.9972e-5(-)	1.1096e-1(+)	6.8570e-2(+)	4.9245e-2
		10	6.0352e+1(-)	7.6654e-1(-)	5.4157e-2(+)	1.3190e+2(-)	1.2858e-1	0.000e+0(-)	2.1800e-5(-)	7.7401e-2(+)	0.000e+0(-)	1.8266e-2
F8	IMA	4	1.0860e+0(-)	4.8780e-1(-)	4.4780e-1(-)	3.4212e-1(=)	3.0474e-1	6.4212e-2(-)	1.5755e-1(-)	1.8169e-1(=)	1.9038e-1(=)	1.8815e-1
		5	4.1261e-1(+)	3.7878e-1(+)	7.2663e-1(-)	4.4856e-1(=)	4.6898e-1	8.0248e-2(-)	9.6311e-2(=)	8.3634e-2(-)	6.6571e-2(-)	9.3411e-2
		6	4.6510e-1(+)	6.2568e-1(-)	6.1188e-1(-)	4.5023e-1(+)	5.2822e-1	5.5515e-2(-)	3.2782e-2(-)	7.8790e-2(+)	4.6611e-2(-)	7.0212e-2
		8	1.4234e+0(-)	6.2627e-1(+)	7.4219e-1(=)	6.7459e-1(+)	7.6233e-1	3.4847e-3(-)	1.0065e-2(-)	3.3662e-2(+)	1.0566e-2(-)	2.8678e-2
		10	2.2724e+0(-)	1.2318e+0(=)	8.6632e-1(+)	8.5436e-1(+)	1.1140e+0	7.985e-10(-)	3.7526e-3(-)	1.3500e-2(+)	1.6578e-3(=)	6.7895e-3
F14	IMA	4	1.4190e+1(-)	1.1848e+1(-)	1.2145e+0(-)	8.5814e+0(-)	1.0579e+0	0.000e+0(-)	0.000e+0(-)	3.5582e-2(+)	0.000e+0(-)	2.6328e-2
		5	1.9367e+1(-)	2.1445e+1(-)	1.8919e+0(-)	1.8854e+1(-)	1.8325e+0	0.000e+0(-)	0.000e+0(-)	4.3356e-2(+)	0.000e+0(-)	1.6210e-2
		6	3.5879e+1(-)	4.6932e+4(-)	1.295e+0(+)	3.5207e+4(-)	1.4040e+0	0.000e+0(-)	0.000e+0(-)	4.2863e-2(-)	0.000e+0(-)	5.4476e-2
		8	2.5594e+1(-)	2.2278e+4(-)	3.9205e+0(-)	2.8626e+4(-)	3.8199e+0	0.000e+0(-)	0.000e+0(-)	0.0000e+0(-)	0.000e+0(-)	0.000e+0
		10	1.8547e+3(-)	3.0434e+4(-)	5.3072e+0(=)	4.5770e+4(-)	5.2711e+0	0.000e+0(-)	0.000e+0(-)	0.0000e+0(-)	0.000e+0(-)	0.000e+0
F15	IMA	4	2.5693e+0(-)	4.4823e+0(-)	6.6938e-1(+)	4.1479e+0(-)	7.9263e-1	0.000e+0(-)	0.000e+0(-)	9.0022e-2(+)	0.000e+0(-)	6.6964e-3
		5	8.7429e+0(-)	4.2029e+1(-)	9.5321e-1(+)	3.4699e+1(-)	1.1204e+0	0.000e+0(-)	0.000e+0(-)	9.7942e-3(-)	0.000e+0(-)	1.7089e-2
		6	1.8817e+1(-)	2.9823e+1(-)	1.4007e+0(-)	1.1066e+1(-)	1.3499e+0	0.000e+0(-)	0.000e+0(-)	2.9979e-4(-)	0.000e+0(-)	1.0634e-3
		8	3.4611e+1(-)	2.8698e+1(-)	2.4502e+0(-)	4.7704e+1(-)	1.1851e+0	0.000e+0(-)	0.000e+0(-)	0.0000e+0(-)	0.000e+0(-)	2.5594e-4
		10	3.2629e+1(-)	7.8776e+1(-)	6.2968e+0(-)	3.2357e+1(-)	2.9317e+0	0.000e+0(-)	0.000e+0(-)	0.0000e+0(-)	0.000e+0(-)	0.000e+0
+/-/=			6/31/8	2/37/6	8/27/10	7/29/9		9/32/4	5/34/6	18/20/7	5/35/5	

D in multimodal and concave problems, it excels in IM and IIGD metrics.

Additionally, it demonstrates robust performance across linear and convex problems. Moreover, in the large-scale partially separable problems as IMAF14–15, IMaOEA-DRA outperforms the comparison algorithms, showcasing its effectiveness and adaptability.

Fig. 12 presents the dominance distribution of various algorithms after performing Bayesian testing on 15,000 samples, without the inclusion of Rope. From the distribution shown in the figure, it is clear that the proposed algorithm outperforms the comparison algorithms across multiple performance metrics, especially in the nine IMAF test problems, where it demonstrates significant performance advantages.

Firstly, from the perspective of the IGD metric, the proposed algorithm exhibits a more precise Pareto front approximation compared to the comparison algorithms. In terms of IHV, the proposed algorithm generates a broader solution space in most problems, covering a wider range of the objective space. This indicates that it not only provides accurate solutions but also ensures diversity and coverage in the solution set, which is particularly important for solving complex optimization problems with multiple conflicting objectives. Similarly, in the IM

metric, the proposed algorithm holds a strong advantage across all nine IMAF test problems. Especially when confronted with optimization problems of higher complexity and changing objectives, the proposed algorithm effectively addresses uncertainties and challenges through its efficient exploration and exploitation strategies.

In summary, the IMaOEA-DRA demonstrates exceptional performance across all three test suites. Notably, it excels in convergence, diversity, and uncertainty when addressing problems with a small number of objectives (fewer than 6), outperforming other contemporary methods. For problems with a larger number of objectives (greater than 6), IMaOEA-DRA exhibits superior uncertainty handling compared to the state-of-the-art IMOEA/D, despite potentially sacrificing some convergence performance on deceptive and degenerate problems. Nonetheless, it retains significant advantages in both convergence and diversity when compared to other classical algorithms.

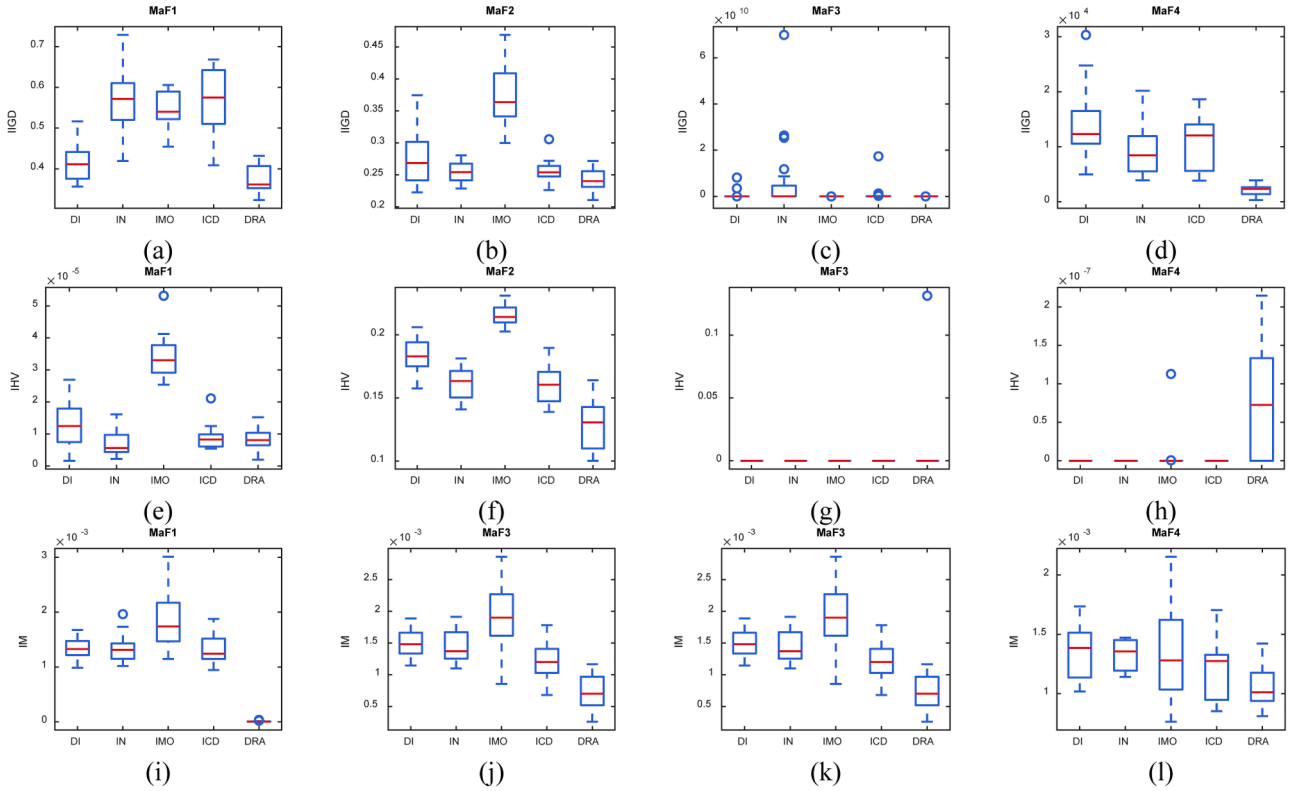
5. Conclusion

In this paper, we proposed an interval many-objective evolutionary

Table 18

IM (mean) obtained across IMAF1–6, 8, 14, 15.

M	Pro.	DI- μ MOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA	Pro.	DI- μ MOGA	IN-MaOEA	IMOEA/D	IMOEA-ICD	IMaOEA-DRA
4	IMa	1.0244e+0(-)	7.2693e-1(-)	1.3102e+0(-)	6.6428e-1(-)	8.8335e-3	IMa	3.1879e-1(-)	4.7135e-2(+)	1.3067e+0(-)	5.3149e-1(-)	1.9861e-1
5	F1	3.4369e-1(-)	2.4364e-1(-)	4.2961e-1(-)	2.4458e-1(-)	1.2504e-2	F6	2.3523e-1(-)	1.2003e-1(-)	4.2465e-1(-)	6.8779e-2(-)	3.5373e-2
6		1.1459e-1(-)	1.0064e-1(-)	1.5003e-1(-)	9.0555e-2(-)	1.2376e-3		4.2194e-2(-)	1.5468e-2(-)	1.2692e-1(-)	5.8557e-2(-)	4.7466e-3
8		1.0387e-2(-)	1.6014e-2(-)	1.5996e-2(-)	1.4297e-2(-)	1.2412e-3		1.0260e-2(-)	6.0837e-3(-)	1.4000e-2(-)	4.1241e-3(-)	3.4740e-5
10		1.5233e-3(-)	1.3880e-3(-)	1.7998e-3(-)	1.1104e-3(-)	1.2484e-5		1.2355e-3(-)	8.7738e-4(-)	1.4296e-3(-)	1.2372e-3(-)	1.1227e-5
4	IMa	8.9410e-1(-)	4.2360e-1(-)	1.3485e+0(-)	6.2399e-1(-)	6.8020e-2	IMa	1.2628e+0(-)	8.7224e-1(-)	1.4900e+0(-)	9.2129e-1(-)	1.4062e-1
5	F2	3.6131e-1(-)	3.2717e-1(-)	4.6898e-1(-)	2.6805e-1(-)	2.5298e-2	F8	3.0146e-1(-)	2.6839e-1(-)	4.2457e-1(-)	2.8469e-1(-)	1.5435e-2
6		1.2755e-1(-)	8.2366e-2(-)	1.5202e-1(-)	1.0485e-1(-)	2.9197e-3		1.1766e-1(-)	1.2920e-1(-)	1.6957e-1(-)	1.0199e-1(-)	1.1525e-2
8		1.4900e-2(-)	1.3791e-2(-)	1.7421e-2(-)	1.3720e-2(-)	7.5050e-4		8.1144e-3(-)	9.6739e-3(-)	1.4607e-2(-)	1.0082e-2(-)	9.0296e-4
10		2.0267e-3(-)	1.3956e-3(-)	2.0160e-3(-)	1.1249e-3(-)	1.5033e-5		1.4433e-3(-)	1.0554e-3(-)	2.0377e-3(-)	1.1799e-3(-)	1.6278e-5
4	IMa	1.7842e-1(-)	2.5292e-1(-)	1.3707e+0(-)	8.5692e-2(-)	2.1569e-2	IMa	8.5514e-1(-)	6.4128e-1(-)	1.1204e+0(-)	2.0165e-1(+)	3.2129e-1
5	F3	2.0970e-1(-)	1.0397e-1(-)	4.2644e-1(-)	3.6959e-2(-)	1.5763e-2	F14	2.9282e-1(-)	1.8245e-1(-)	3.7506e-1(-)	1.7489e-1(+)	2.0512e-1
6		8.8987e-2(-)	6.7242e-2(-)	1.5335e-1(-)	6.9878e-2(-)	5.5603e-3		9.2179e-2(-)	5.8862e-2(+)	1.3503e-1(-)	3.7076e-2(+)	6.9067e-2
8		1.2060e-2(-)	1.0641e-2(-)	1.7102e-2(-)	1.1327e-2(-)	3.0035e-4		1.3473e-2(-)	8.2516e-3(-)	1.3927e-2(-)	9.4220e-3(=)	7.1085e-3
10		1.2526e-3(-)	1.2590e-3(-)	1.8499e-3(-)	1.3086e-3(-)	1.3784e-6		1.2663e-3(-)	1.6585e-3(-)	1.2759e-3(-)	1.4634e-3(=)	6.0115e-4
4	IMa	7.8393e-1(-)	1.0203e+0(-)	1.1324e+0(-)	7.4730e-1(-)	1.2674e-1	IMa	8.4102e-1(-)	5.5792e-1(-)	1.5512e+0(-)	5.5896e-1(-)	3.8015e-1
5	F4	3.5380e-1(-)	3.6571e-1(-)	3.8694e-1(-)	2.8743e-1(-)	5.0355e-2	F15	2.9377e-1(-)	2.5550e-1(-)	5.2968e-1(-)	3.6066e-1(-)	1.3342e-1
6		9.2320e-2(-)	8.5602e-2(-)	1.2464e-1(-)	1.0578e-1(-)	1.2088e-2		1.2800e-1(-)	1.3939e-1(-)	1.5319e-1(-)	1.1089e-1(-)	5.8031e-2
8		1.1913e-2(-)	9.7832e-3(-)	1.3758e-2(-)	1.0073e-2(-)	2.8273e-3		1.3874e-2(-)	1.4583e-2(-)	1.7942e-2(-)	1.3150e-2(-)	3.8525e-3
10		1.2003e-3(-)	1.4145e-3(-)	1.3418e-3(-)	1.3849e-3(-)	1.0569e-3		1.5385e-3(-)	1.0218e-3(=)	1.7791e-3(-)	1.2845e-3(-)	1.0279e-3
4	IMa	1.2087e+0(-)	6.5685e-1(-)	1.3315e+0(-)	8.7733e-1(-)	8.9244e-2						
5	F5	3.4846e-1(-)	3.0718e-1(-)	4.3299e-1(-)	1.2498e-1(-)	5.0142e-2						
6		1.1595e-1(-)	6.6121e-2(-)	1.3392e-1(-)	8.2116e-2(-)	2.3486e-2						
8		8.8273e-3(-)	1.1636e-2(-)	1.5150e-2(-)	1.0768e-2(-)	2.3637e-3						
10		1.0736e-3(-)	1.7692e-3(-)	1.6924e-3(-)	2.0083e-3(-)	1.9755e-4						
+/-/=		0/45/0					0/45/0		3/40/2			

**Fig. 11.** Box-plot of IIGD, IHV, IM with 10 objectives on the IMAF1–4.

algorithm based on dynamic relation adjustment named IMaOEA-DRA as a novel approach for addressing interval many-objective optimization problems. IMaOEA-DRA dynamically adjusted the convergence, uncertainty, and diversity of the interval individual evolution process through an iterative procedure, resulting in improved solutions. Specifically, to guide the evolution direction of interval individuals, we designed a new interval dominance relation called DIP-dominance that incorporates a predefined decay function. Furthermore, to maintain the

diversity of the interval population, a new interval evolution indicator named PIBI indicator and an interval clustering strategy were proposed.

To sum up, our approach partially relaxes the Pareto dominance relation, rendering it applicable to interval problems. Finally, we evaluated the performance of IMaOEA-DRA on 23 benchmark sets for IMaOPs and our method demonstrates outstanding performance.

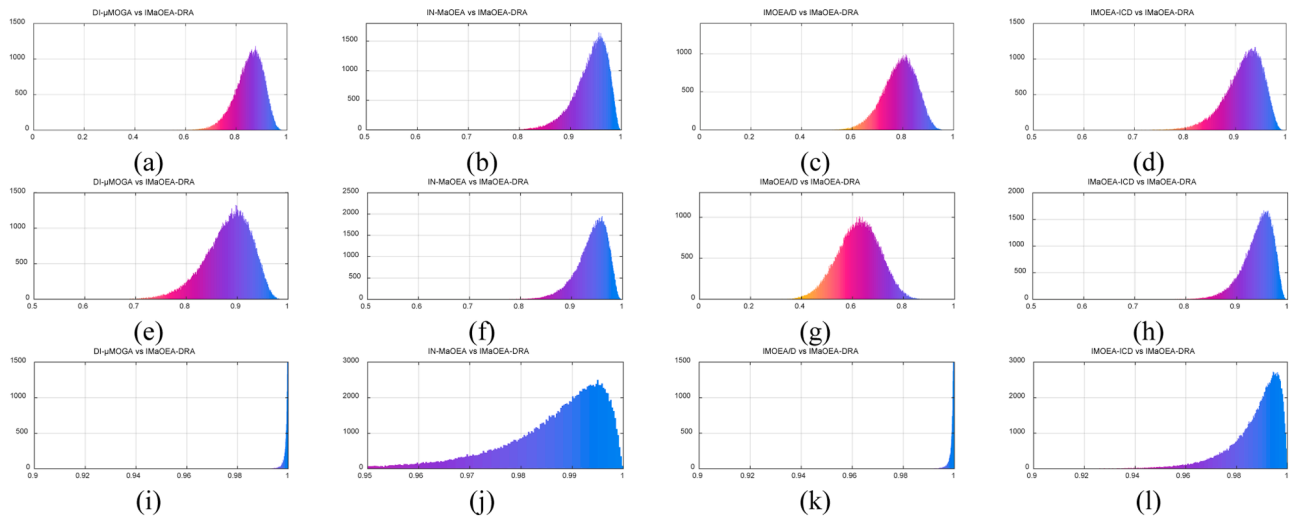


Fig. 12. Sample distribution across the four algorithms in the Bayesian test by IMAf test suites. (a)-(d): IIGD, (e)-(h): IHV, (i)-(l): IM.

CRedit authorship contribution statement

Zhang jingbo: Writing – original draft. **Zhang zhixia:** Conceptualization. **Cai xingjuan:** Funding acquisition. **Cai jianghui:** Investigation. **Chen jinjun:** Investigation.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication.

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Data availability

Data will be made available on request.

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